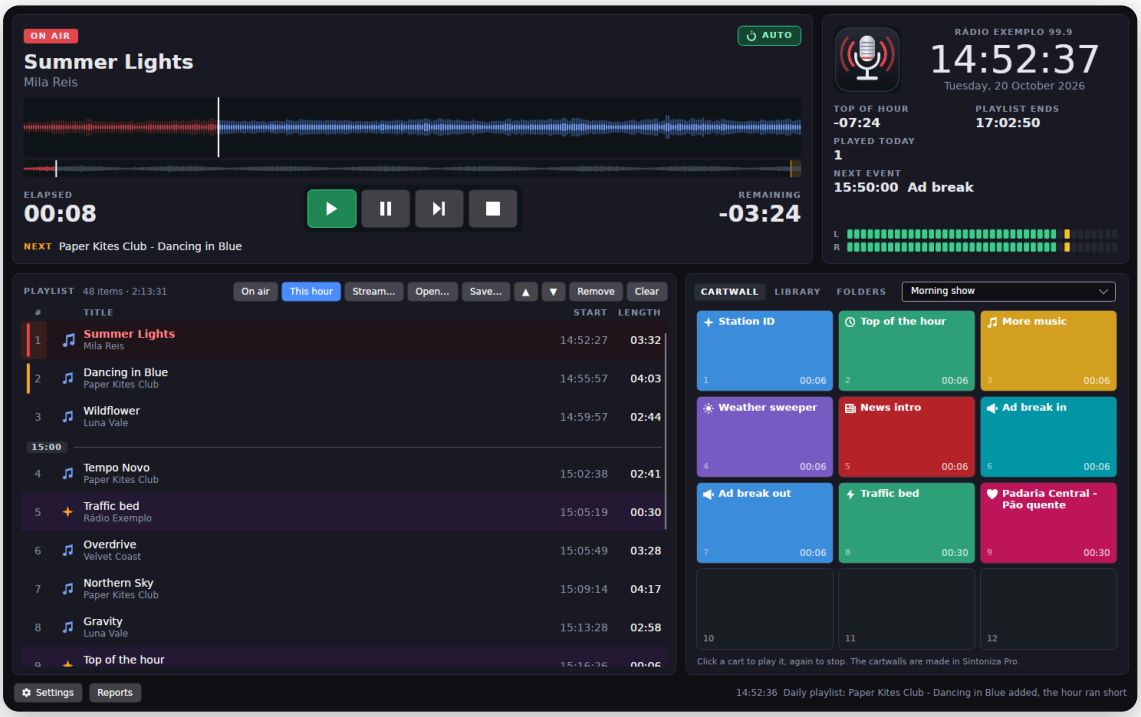
Sintoniza Radio Broadcast system

User manual



Version v1.1.20-6-g579f9e0 · 20 October 2026 · cesararaujo.net

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1 Sintoniza at a glance

Sintoniza is radio automation for stations that are on air around the clock. It is made of three programs that share one database, the **library**: the music and its categories, the schedule, the daily playlists and the history of everything that went on air.



Sintoniza Radio Broadcast system

The on-air program, in the studio: the playlist, the cartwall, the microphone and the stream. The host uses it.



Sintoniza Pro

The station's production program: its settings, categories and cartwalls; events, ad breaks, clients and campaigns, hour templates for your shows, and the daily playlists song by song.



Sintoniza Import

Brings folders of music into the library, in categories.

Each has its own entry in the app menu (Linux), the Start menu and desktop (Windows) or Applications (macOS), and only one copy of each runs on a computer. With a shared MySQL / MariaDB library, Pro and Import can run on other computers (the commercial department, a music office); the studio sees their changes within seconds.

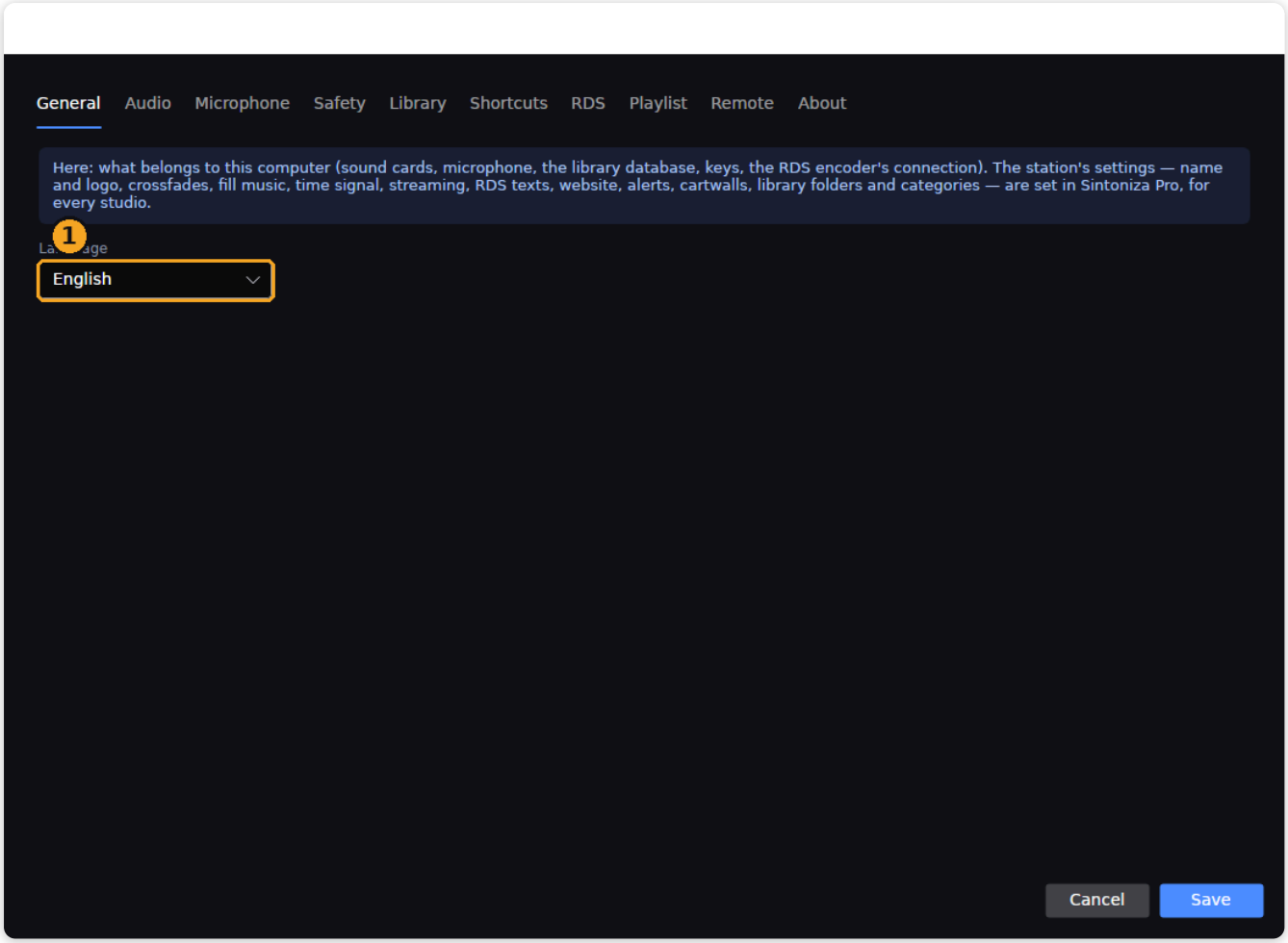
> **TIP**
>
> The on-air program's **Settings** only has what belongs to that computer: sound cards, microphone, the database, keys. **Everything about the station** — name and logo, crossfades, fill music, time signal, streaming, RDS texts, website, alerts, categories, cartwalls — **is set once in Sintoniza Pro**, for every studio.

How the day is planned and played

1. **Import** the music into categories, under the station's eight (Music › Current, Gold, 80s, Jingles, Spots…).
2. In **Pro**, make **hour templates** for your shows (which categories play, in which order) and place them on the week; book the **ad breaks** and their **clients**.
3. The **daily playlists** are made from the templates for today and the next days. Look at them, change songs by hand, or let the rules fill them.
4. The **on-air program** plays the daily playlists, starts the ad breaks and events at their times, and keeps every hour on time.

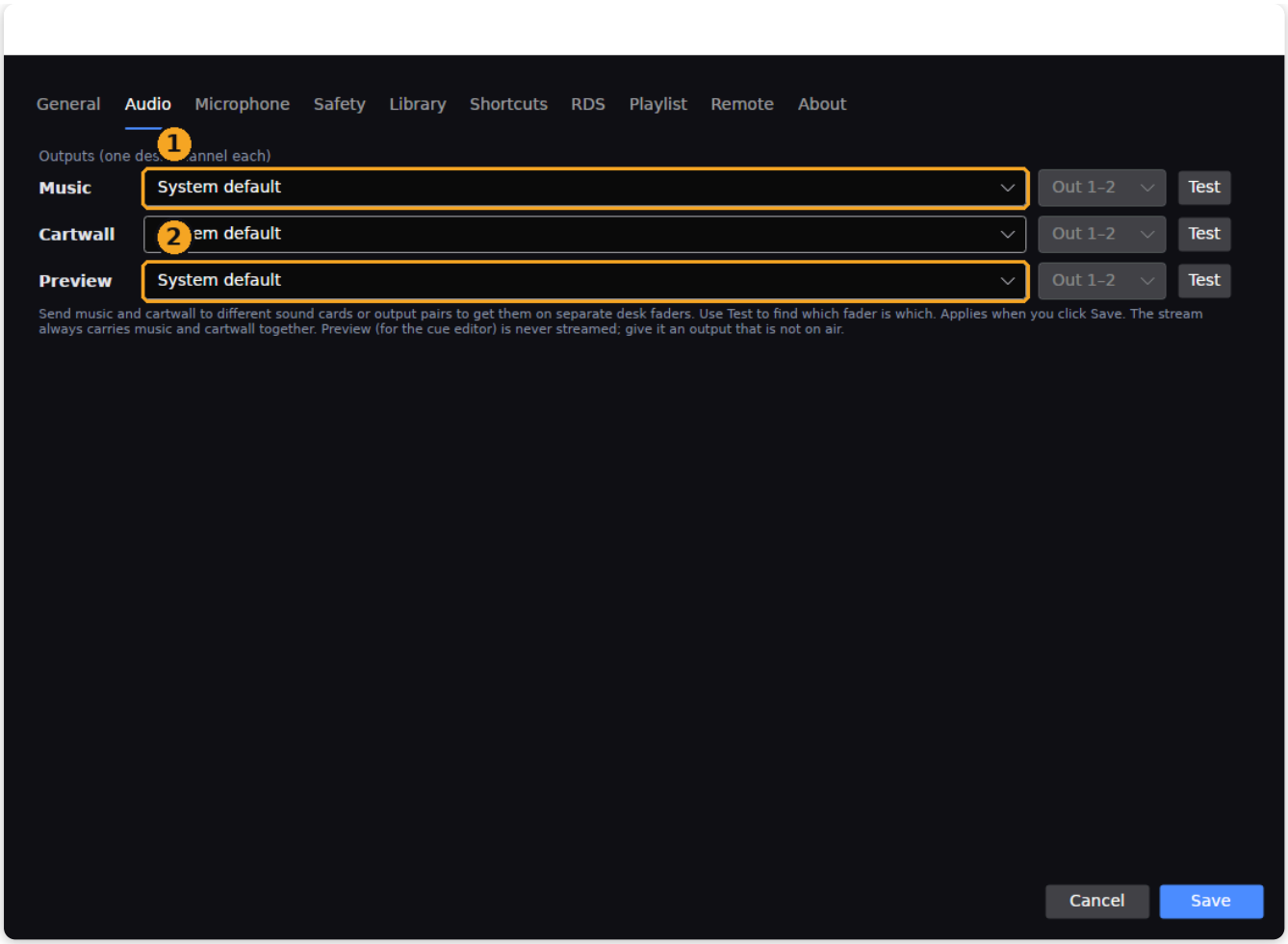
2 First setup

On **each studio computer**, in the on-air program: click **Settings** (bottom left, or Ctrl/Cmd + ,). Only what belongs to that computer is there. **Save** applies the changes.



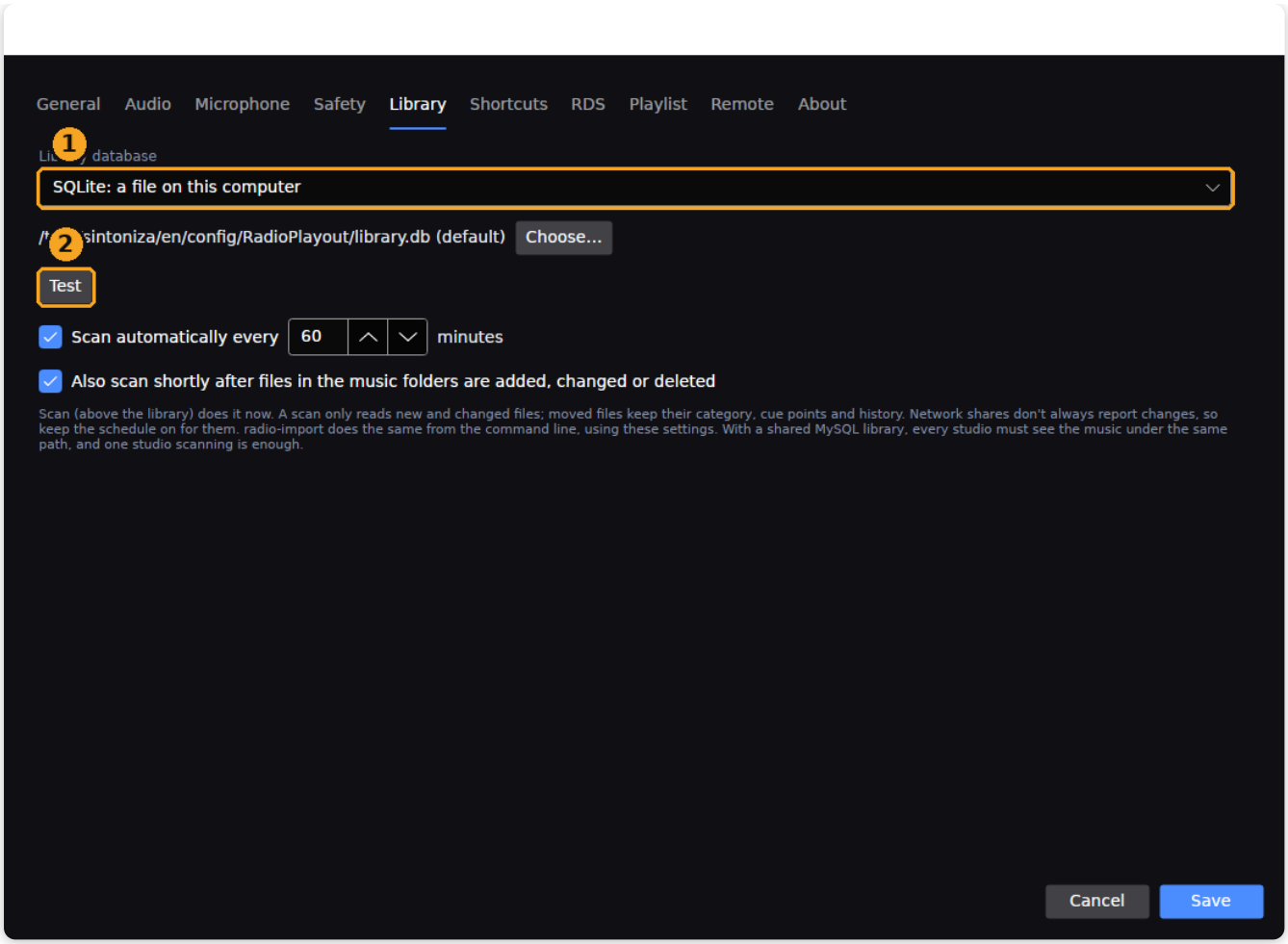
Settings → General

1. The **language** of the screens (English, Português, Français); it applies the next time the program starts.



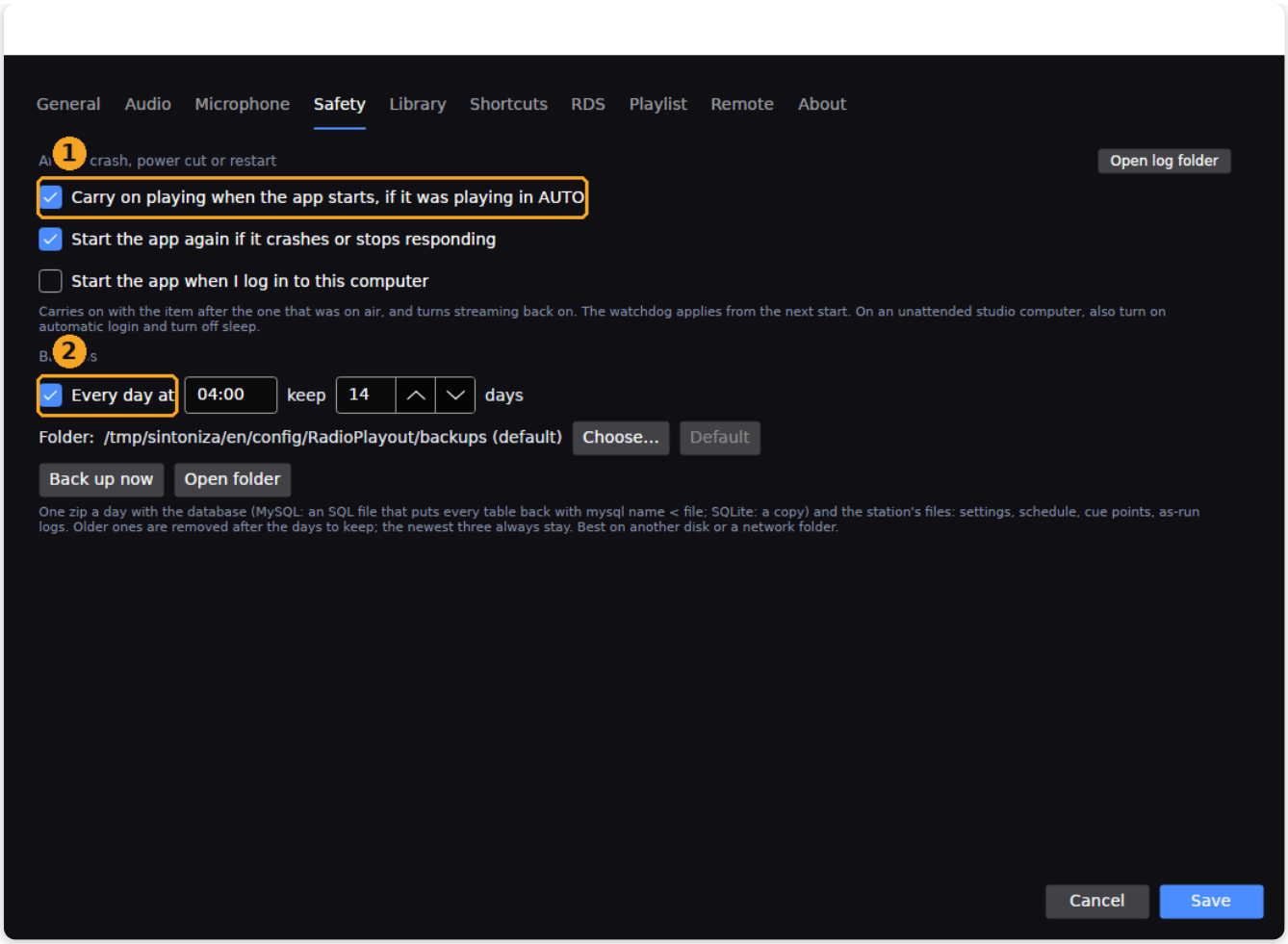
Settings → Audio

1. The sound card and channel pair for **Music**; Cartwall and Preview below it. **Test** plays a short tone.
2. **Preview** is for listening before air: headphones or the desk's PFL/cue, never an on-air output.



Settings → Library

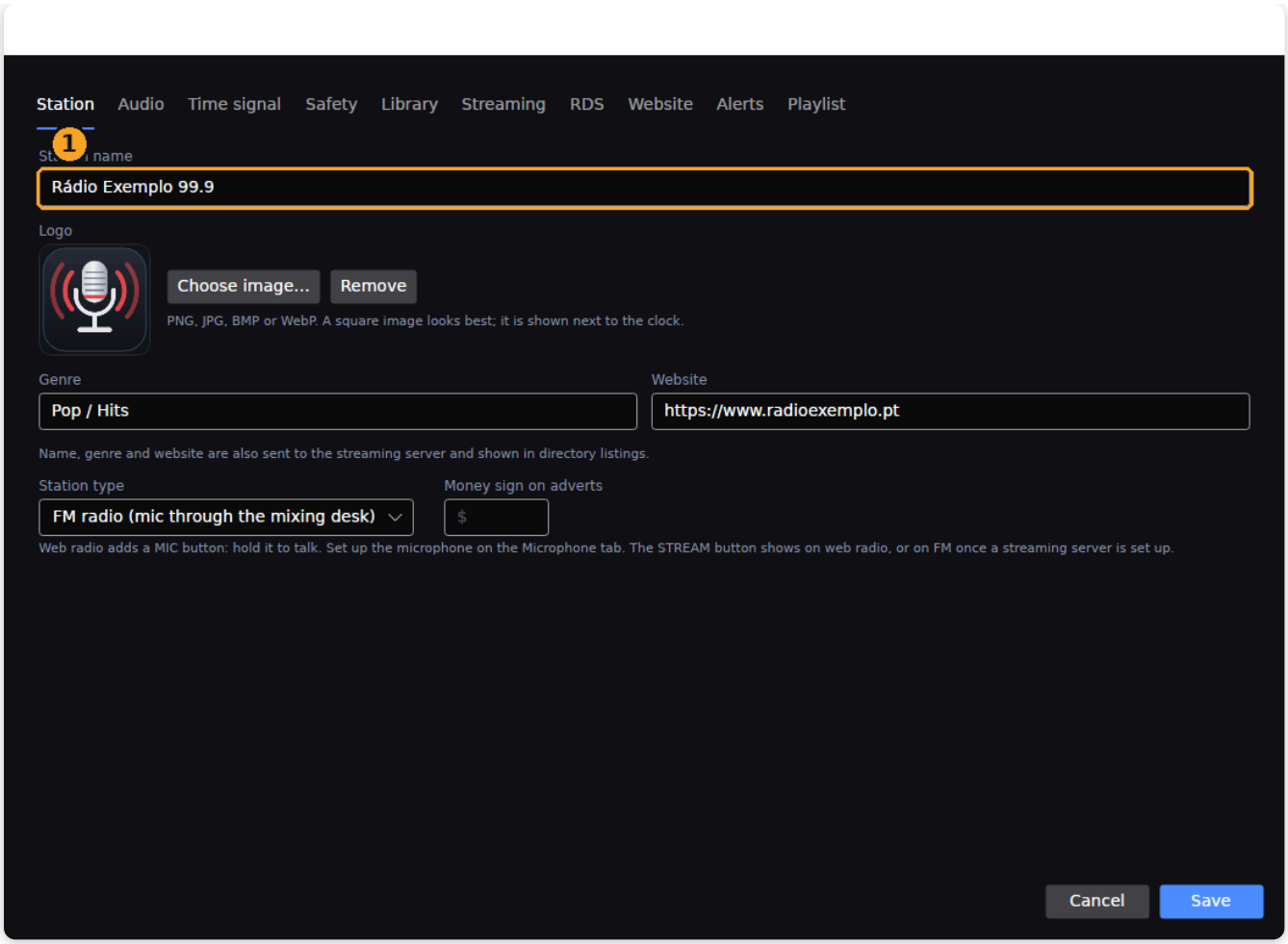
1. Where the library lives: *SQLite* (a file on this computer) or *MySQL / MariaDB* (a server several studios share).
2. **Test** opens it (and makes its tables the first time).



Settings → Safety

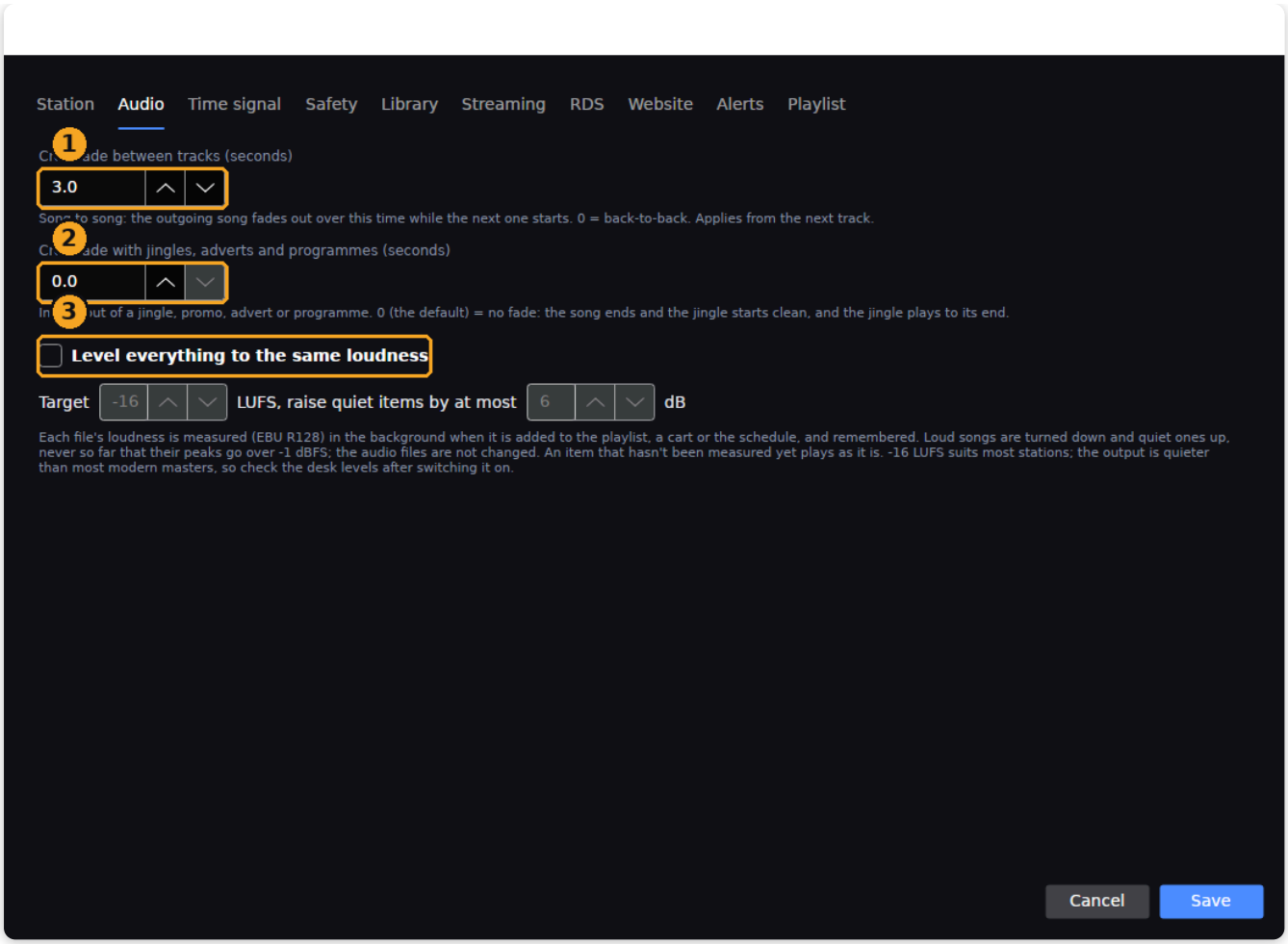
1. **Carry on playing when the app starts**, **Start the app again if it crashes**, **Start the app when I log in**: for a station that is never off air.
2. **Backups**: every day, of the database and the station's files.

Then **once, for the whole station**, in **Sintoniza Pro → Station settings**:



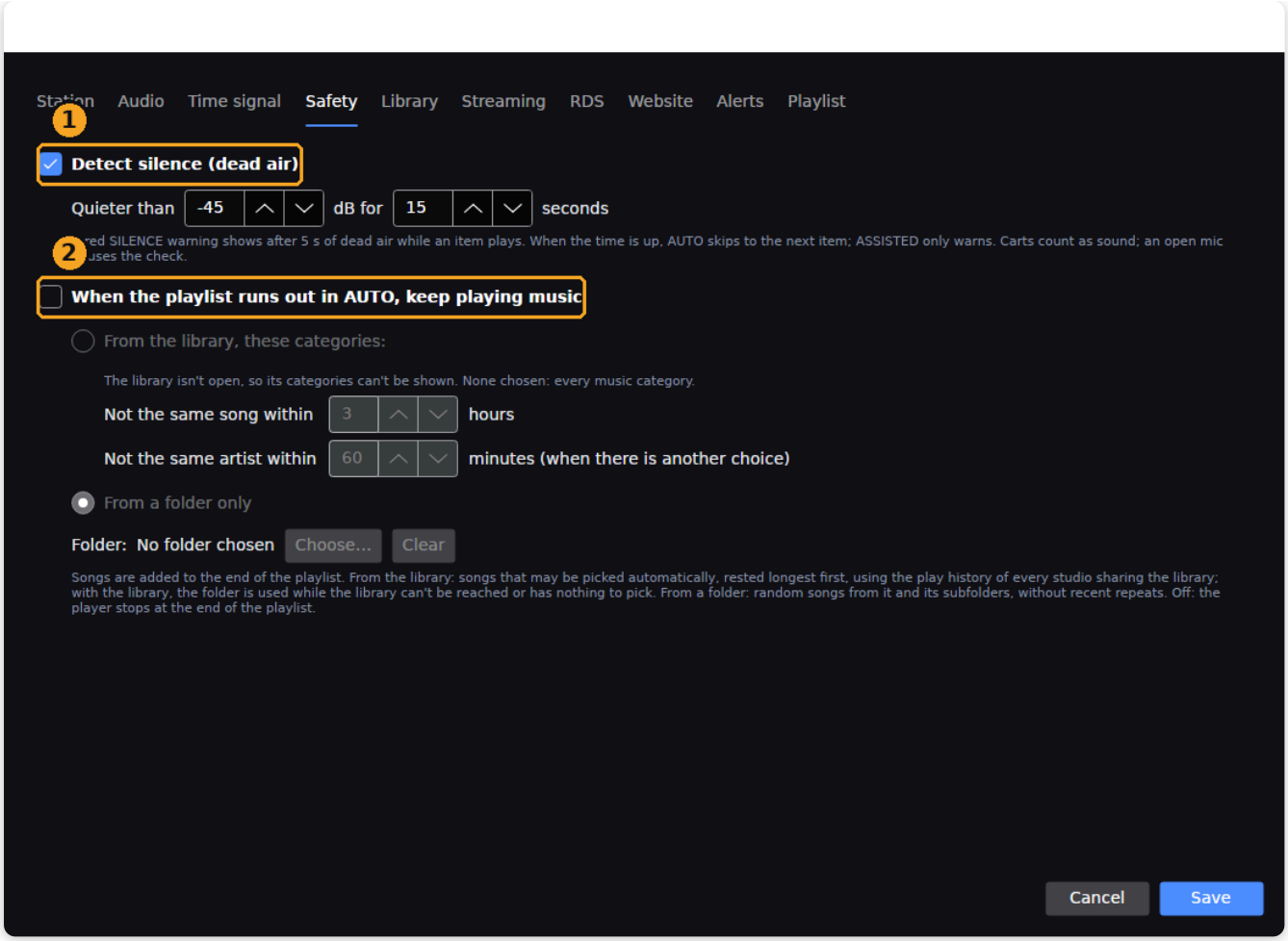
Station settings → Station

1. **Station name**, **logo**, genre and website: shown next to the clock and sent to the stream and RDS. **Station type**: *FM radio* (microphone through the mixing desk) or *Web radio* (through the computer).



Station settings → Audio

1. Crossfade from song to song, in seconds.
2. Crossfade into or out of jingles, adverts and programmes (0 = clean cut).
3. Every item at the same loudness (EBU R128).



Station settings → Safety

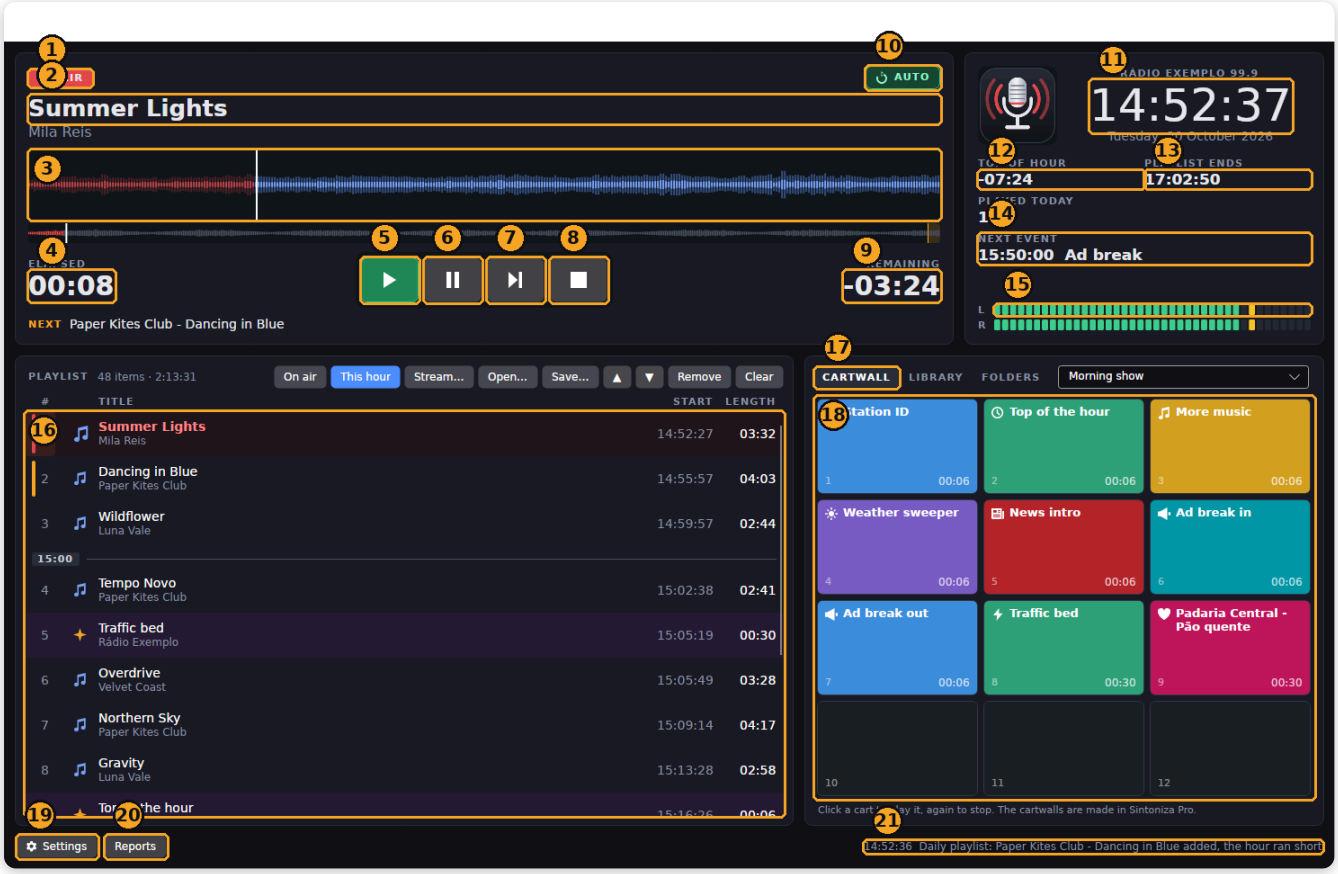
1. **Detect silence**: an alarm (and, in AUTO, a skip) when nothing is heard.
2. **Fill music**: what plays if the playlist ever runs out, and the rest times (a song not again within N hours, an artist not within N minutes). The daily playlists use the same rest times.

The other tabs there: **Time signal**, **Library** (other music folders), **Streaming**, **RDS** (the texts), **Website**, **Alerts** and **Playlist** (days made ahead, row colours). Every studio takes the station's settings within a minute of saving. The first studio started after updating to this version publishes its own.

> **TIP**
>
> Then import the music (chapter 4), make the cartwalls, build the templates and ad breaks (chapter 3), switch the on-air program to **AUTO** and press **Play**.

3 The on-air program

Sintoniza Radio Broadcast system is what the host works with. The top half is what is on air now; the bottom half is what comes next (the playlist) and what is at hand (cartwall, library, folders).

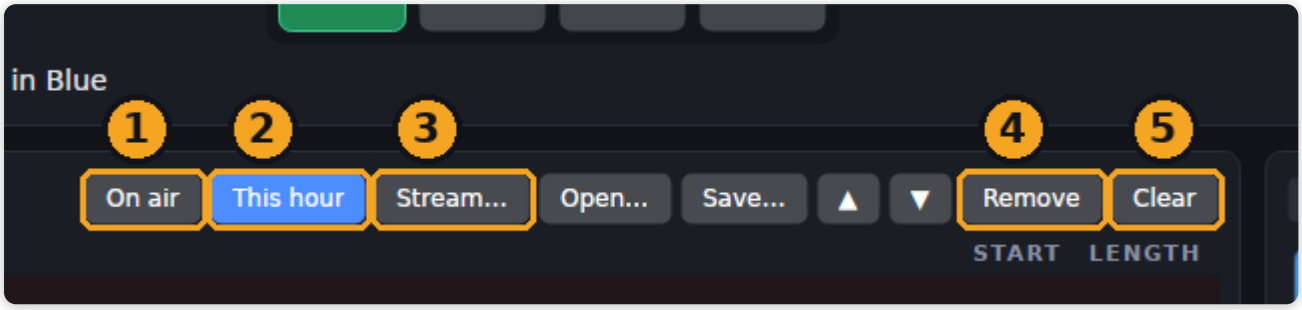


The on-air screen

1. State: **ON AIR**, **STOPPED**, **SILENCE** (nothing heard), or the countdown to the next ad break.
2. What is on air, with its artist.
3. The waveform: the red part has played. The thin bar below is the whole song.
4. Time played, and the **NEXT** item below it.
5. **Play** (F5).
6. **Pause** / resume (F6).
7. **Next** (F7): crossfades to the next item now.
8. **Stop** (F8).
9. Time left of the item on air.
10. **AUTO** runs through the playlist; **ASSISTED** stops after each item until you start the next.
11. The clock, the station name and logo.
12. Countdown to the top of the hour.
13. When the playlist runs out.
14. The next scheduled event or ad break.
15. Output levels, left and right.
16. The playlist (next pages).
17. **CARTWALL**, **LIBRARY** and **FOLDERS** tabs.
18. The cartwall: jingles and sounds at hand.
19. **Settings** (Ctrl/Cmd + ,).
20. **Reports**: music played and proof of broadcast.
21. The last thing that happened (all of it is in the log).

The playlist

Each row shows a type icon (♫ song, ✦ jingle, € advert, 🎙 programme, stream), the title and artist, the expected **start** time and the **length**. Everything that isn't a song has its own background colour (Settings → Playlist). The item on air fills with red as it plays, and a line marks where each hour starts.



Playlist buttons

1. **On air**: jump back to the item on air (it lights up when you scroll away).
2. **This hour**: the hour before, this hour and the first 30 minutes of the next. Off: everything still to play.
3. **Stream…** another station's stream; next to it **Open…** and **Save…** (M3U playlists) and ▲▼ to move. Songs come from the Library on the right.
4. **Remove** the selected items.
5. **Clear**: empties the playlist except the item on air and what the schedule locked; the plan refills it.

- **Double-click** an item (or press Enter) to put it on air now.
- **Drag** rows to reorder them, or drop songs from the library, the folders tab or your file manager.
- **Right-click**: *Edit cue points…* (Ctrl/Cmd+E), *Edit tags…* (Ctrl/Cmd+T), *Remove*.
- Rows tagged **SCHEDULED** are ad breaks and events coming up, shown where they will play.

> **IMPORTANT**
>
> **Adverts are protected.** While an advert is on air, Next, Pause, Stop and putting another item on air ask first. Only a click on the button in the question does it, so a paid spot isn't cut by accident.

Where the songs come from: the daily playlists (chapter 3). About half an hour before each hour, its songs are put in the playlist. Near the end of the hour the program keeps time: if the next hour would start late, an *optional* item is left out; if it would start early, songs are added. You can still change the playlist on air.

Cartwall

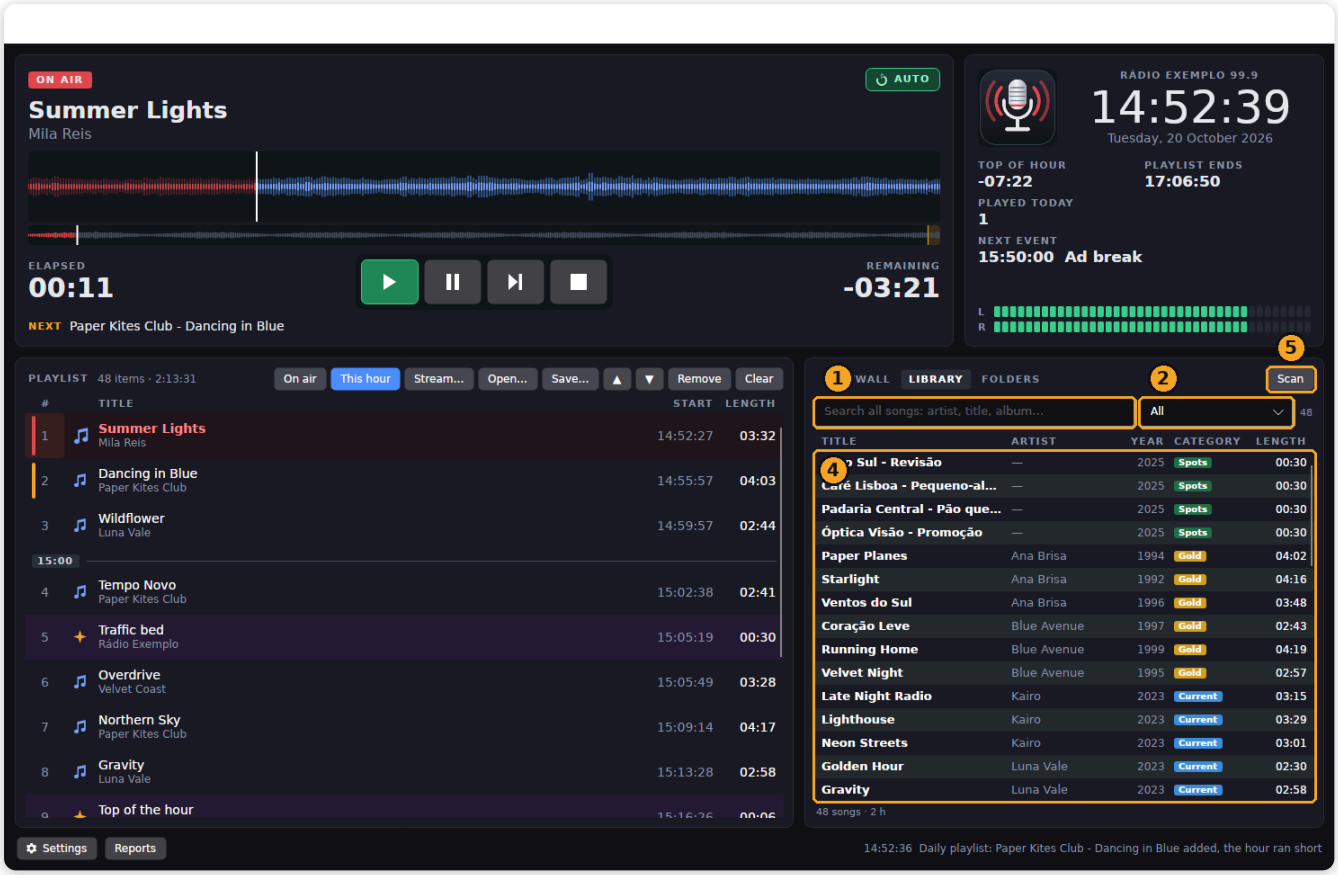


The cartwall

1. The carts: name, icon and colour; the number is the cart's key (Settings → Shortcuts).
2. Which cartwall is shown.

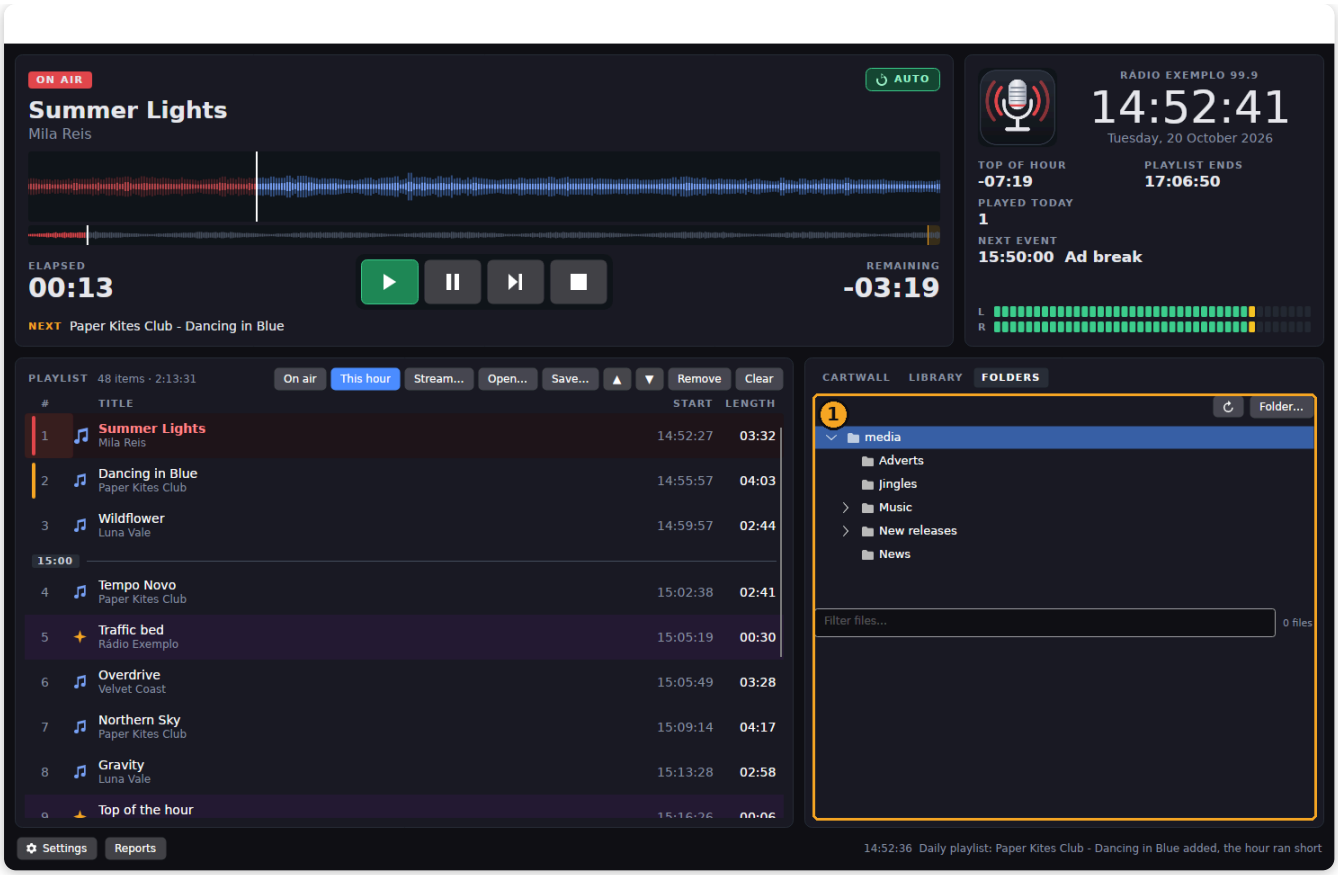
- **Click** a cart to play it, click again to stop. Carts play over the music on the Cartwall output. Each shows how long it lasts; while it plays, the time left.
- The cartwalls are the station's: they are made in Sintoniza Pro → **Cartwalls**, and every studio gets them within a minute.
- Keys for each cart: Settings → Shortcuts.

Library



The Library tab (Ctrl/Cmd + F)

1. Search every song as you type: artist, title, album or file name (case and accents don't matter).
2. Only one category (a category includes the ones under it).
4. The songs. **Double-click** or Enter: to the end of the playlist. **Right-click**: Play next, Category, Also in, Tempo, Voice, Language, Edit tags…, Edit cue points…
5. **Scan**: bring the library up to date with the music folders now.



The Folders tab

1. The music folders on disk: drag files or folders to the playlist. Right-click a folder → *Category* puts all its songs (and songs put there later) in a category.

Cue points



Edit cue points…

Set **Cue In** (where it starts), **Intro** (end of the intro, for talking over it), **Outro**, **Mix** (where the next item starts) and **Cue Out**. *Auto-detect* finds the first and last sound and a good Mix point. Space plays and pauses on the Preview output; ← → move a second (Shift: 0.1 s); the mouse wheel zooms.

Segues

How one song goes into the next: right-click a playlist item → *Segue with the next…* (Ctrl/Cmd+G), or **Segue…** in Daily playlists.

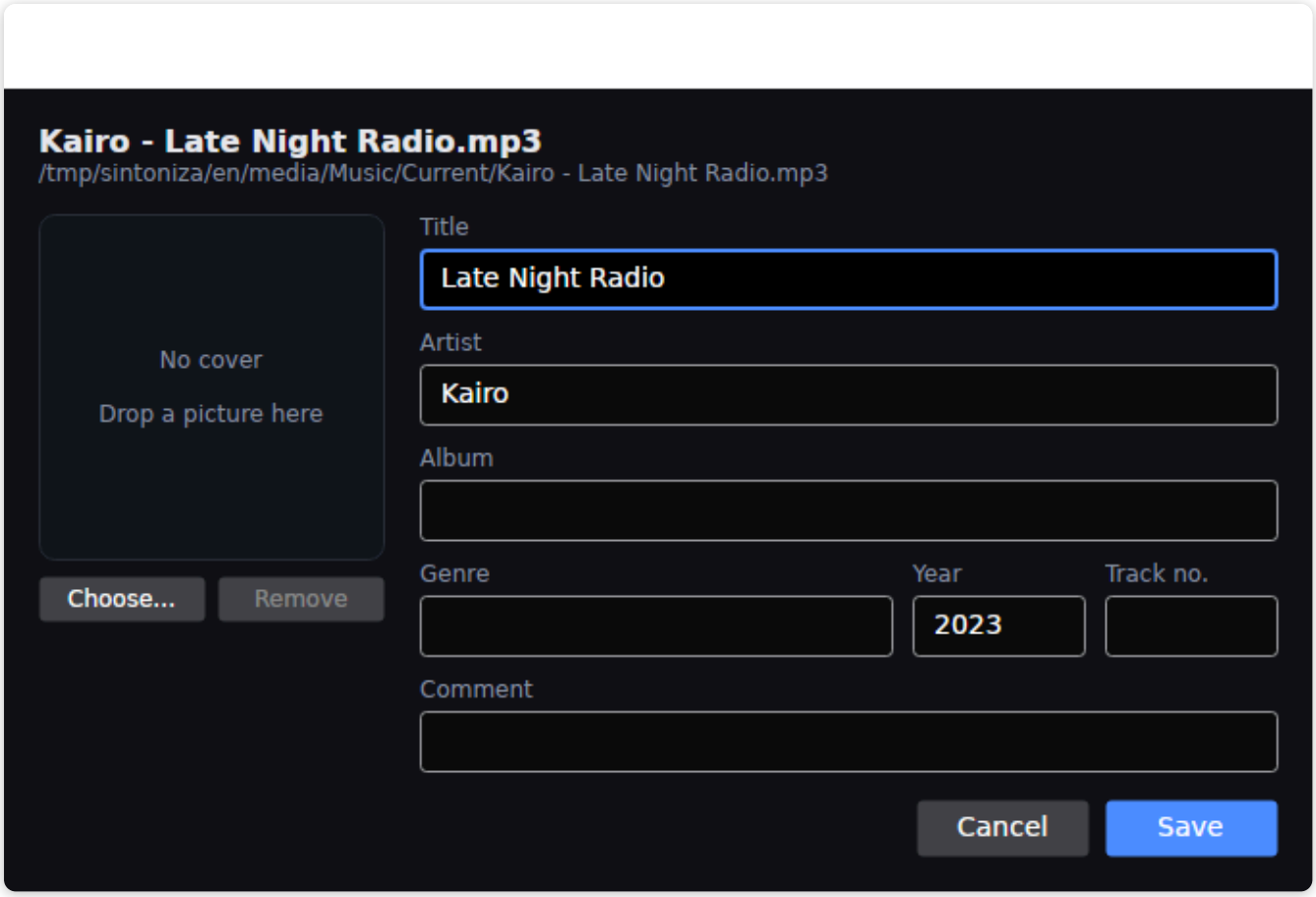


Segue with the next...

1. The end of the first song (top) and the start of the next (bottom), joined at 0. **Drag** either song to move the join; the yellow line is the first one's level (its fade).
2. **Mix**: where the next song starts. Moving it makes a Mix marker: the first song then plays on at full level to its Cue Out. *Automatic* goes back to the usual crossfade.
3. **Cue Out**: where the first song stops.
4. **Cue In**: where the next song starts in its file (not past its Intro).
5. **Play the join** from a few seconds before, exactly as on air, on the Preview output (Space). Change something while it plays and you hear it again at once.
6. What will happen, in words.

Saved as the two songs' cue points: they apply wherever the songs play, in every studio sharing the library.

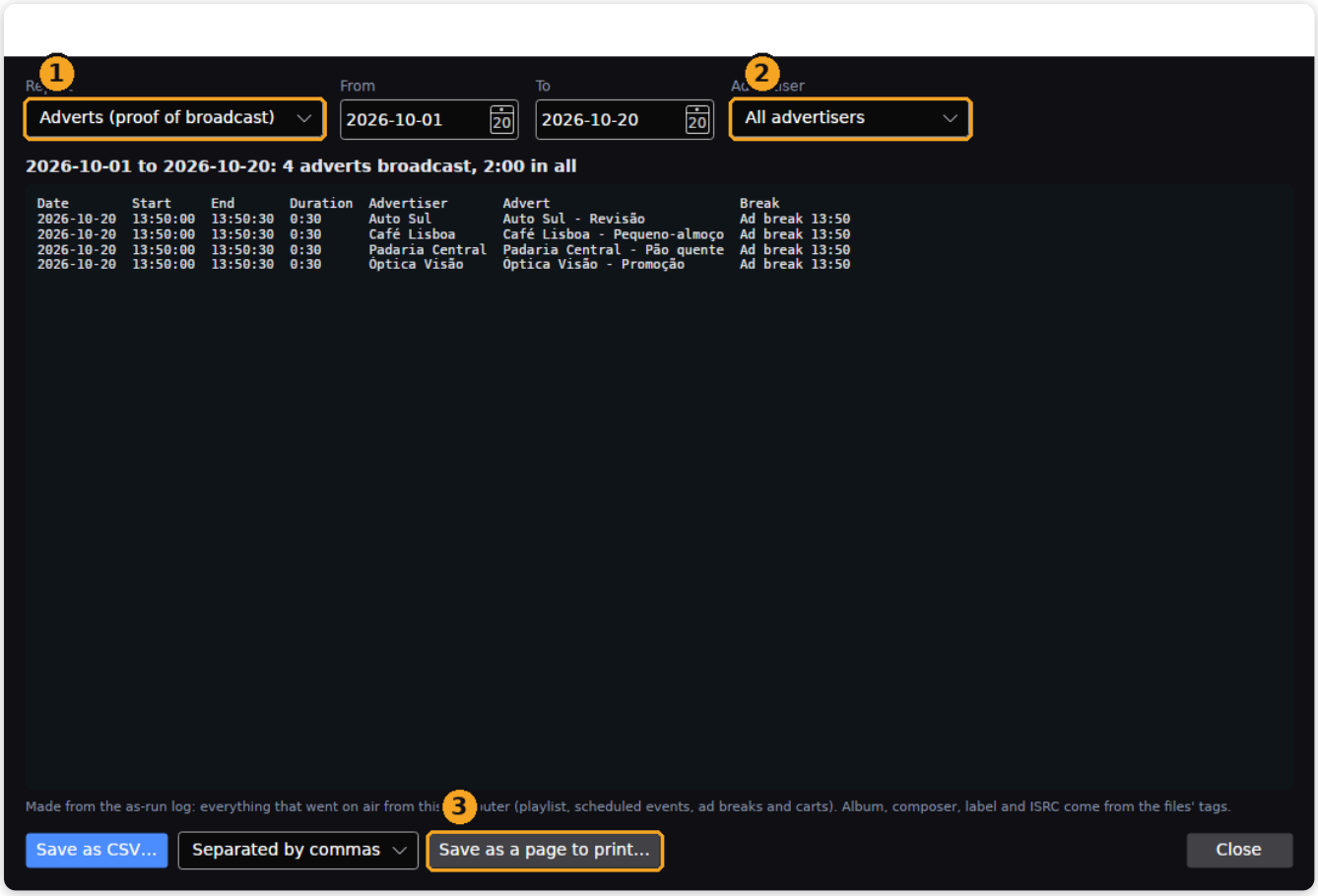
Tags



Edit tags...

Title, artist, album, genre, year, track number, comment and cover picture, written in the file. A file that is on air can't be changed until it ends.

Reports

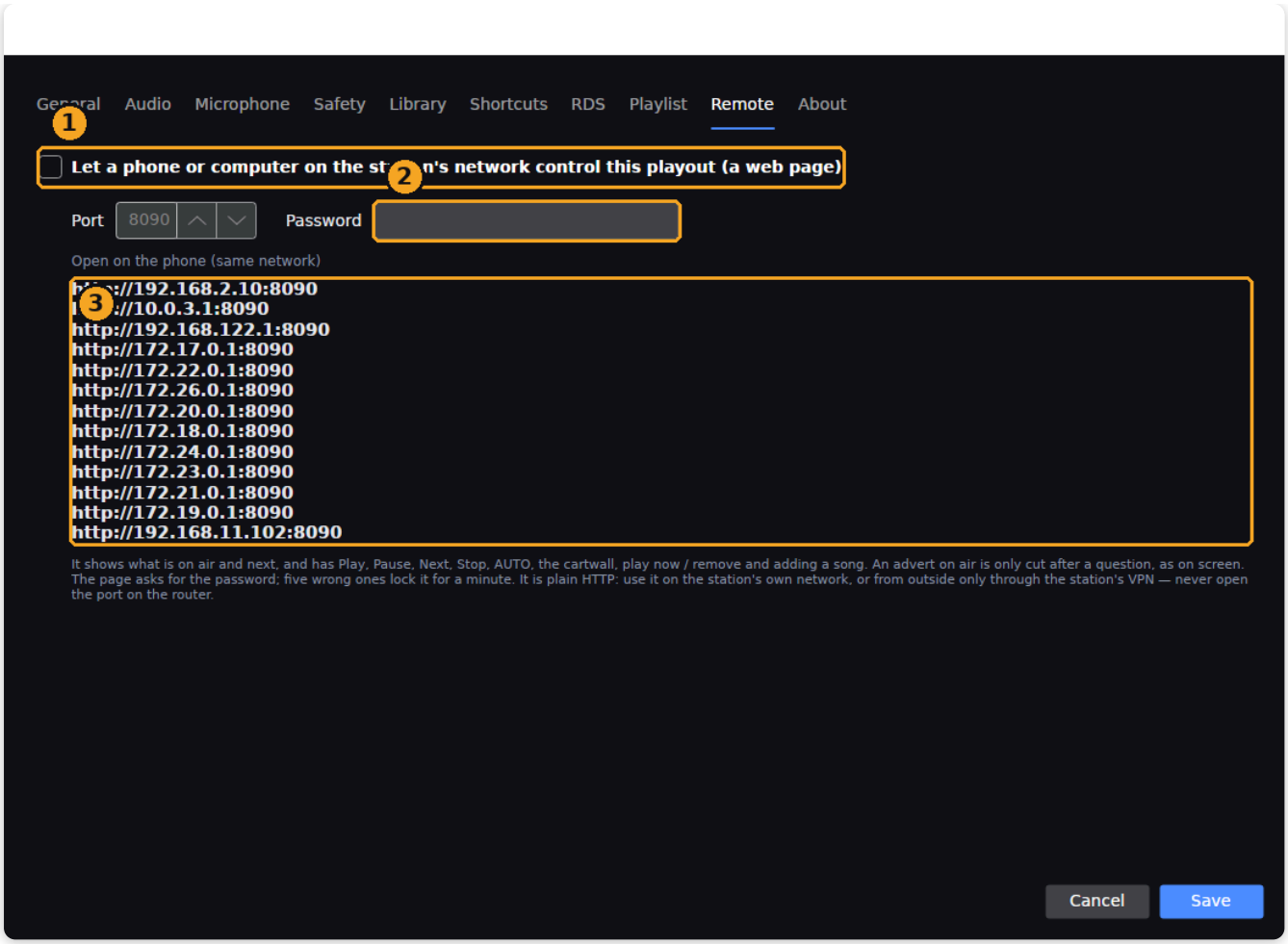


Reports

1. **Music played** (royalties), **Music totals per song**, **Adverts** (proof of broadcast) or **Everything** (as-run log).
2. For the proof of broadcast: one advertiser or all.
3. **Save as a page to print…** (or as PDF); next to it, **Save as CSV…** for a spreadsheet.

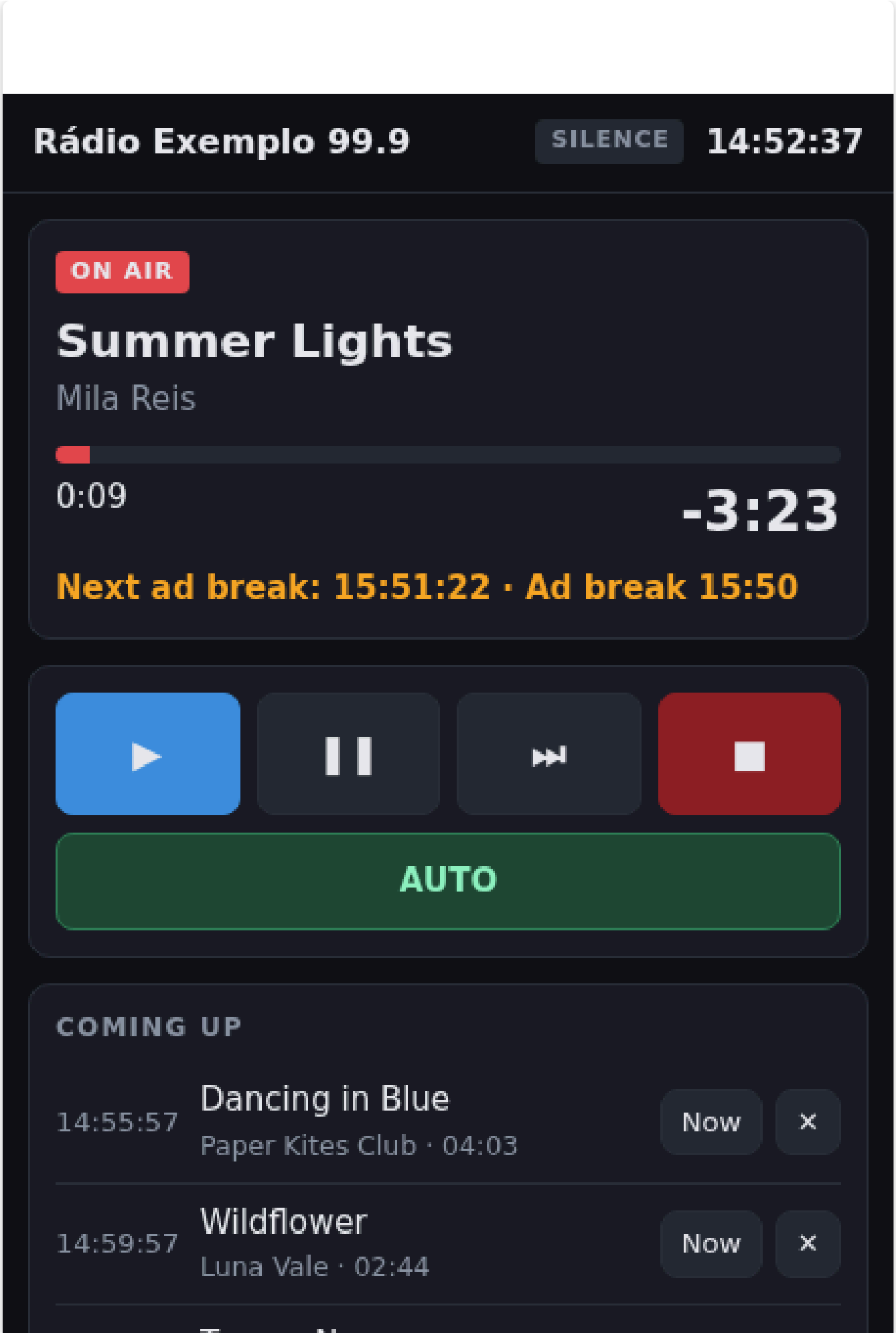
Web remote

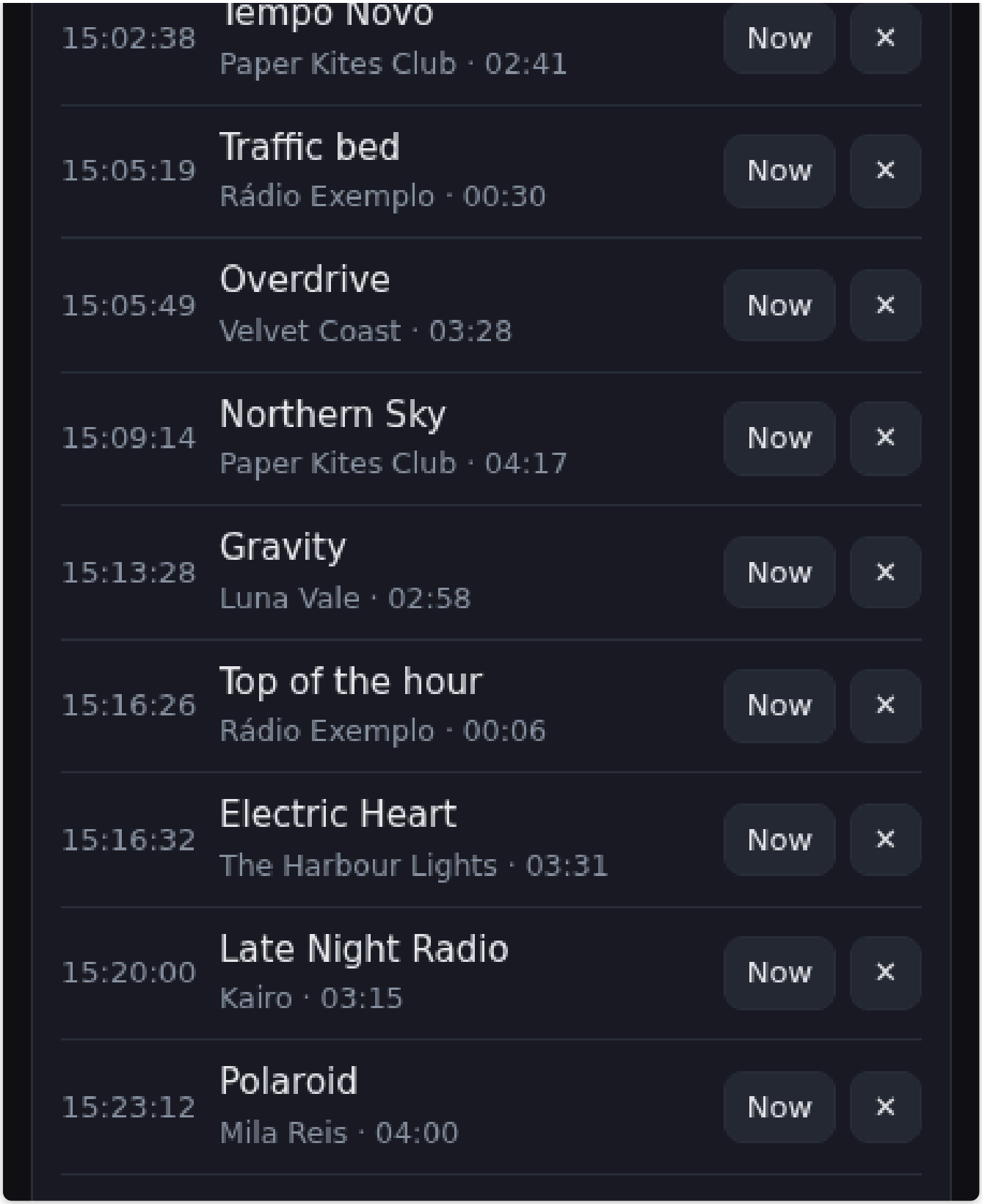
Control the playout from a phone or another computer on the station's network: a web page, no app to install.



Settings → Remote

1. Switch it on (it is off until you do).
2. The **password** the page asks for (and the port).
3. The address to open on the phone, on the same network.


Rádio Exemplo 99.9
SILENCE
14:52:37
ON AIR
Summer Lights
Mila Reis
0:09
-3:23
Next ad break: 15:51:22 · Ad break 15:50
AUTO
COMING UP
14:55:57
Dancing in Blue
Paper Kites Club · 04:03
Now
14:59:57
Wildflower
Luna Vale · 02:44
Now



The web remote on a phone

- What is on air, the time left and the next ad break; lights for the stream and silence.
- ▶ ❙❙ ⏭ ■ and **AUTO**, as on screen. An advert on air is only cut after a question.
- **Coming up**: *Now* puts an item on air, × removes it. The cartwall's carts fire with a tap.
- **Add a song**: search the library and add it after the item on air.

> **IMPORTANT**
>
> It is plain HTTP, for the station's own network. From outside, use it only through the station's VPN — never open its port on the router. Five wrong passwords lock the page for a minute.

The microphone (web radio)

With *Web radio* chosen (Pro → Station settings), the **MIC** button (F9) opens the microphone and the music and carts fade down while it is open: hold to talk, or press on and press off (Settings → Microphone). With *FM radio* the microphone goes through the desk.

All the settings

This computer's, in the on-air program's **Settings**:

Tab	What it is for
General	The language of the screens
Audio	Outputs for Music / Cartwall / Preview
Microphone	(web radio) input, levels, how far the music goes down
Safety	Start-up and crash behaviour, the log, backups
Library	The database, automatic scans
Shortcuts	Keys for Play, Pause, Next, Stop, MIC, the stream and each cart
RDS	Whether this computer sends to the RDS encoder, and how it reaches it
Playlist	How much of the next hour the playlist shows
Remote	The web remote: on/off, port, password, its address
About	Version, updates and this manual

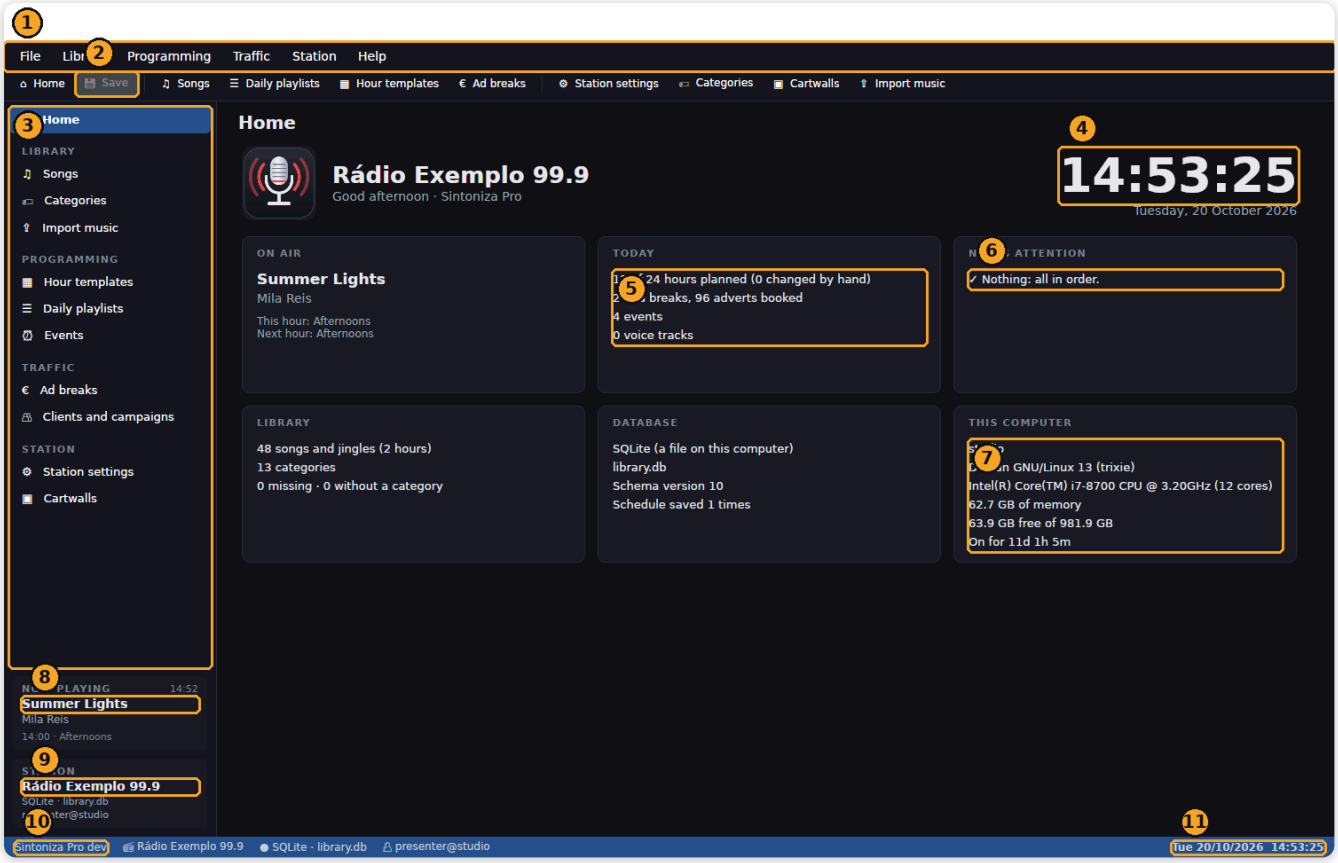
The station's, in **Sintoniza Pro → Station settings**:

Tab	What it is for
Station	Name, logo, genre, website, FM or web radio, money sign
Audio	Crossfades, loudness levelling
Time signal	Pips or a file at the top of the hour; hour announcements (HRS05.mp3 at 05:00…)
Safety	Silence detection, fill music
Library	Other music folders (songs without a category)
Streaming	Icecast / Shoutcast server, format and bitrate

Tab	What it is for
RDS	PS, radio text, RT+
Website	Now playing posted to the station's site
Alerts	Telegram, e-mail or web alerts: silence, stream down, crash, low disk, clock, backups
Playlist	How many days of daily playlists are made ahead, row colours

4 Sintoniza Pro

The station's production program: its settings, library, categories, cartwalls, schedule and advertising. It opens on **Home**, with the station at a glance.



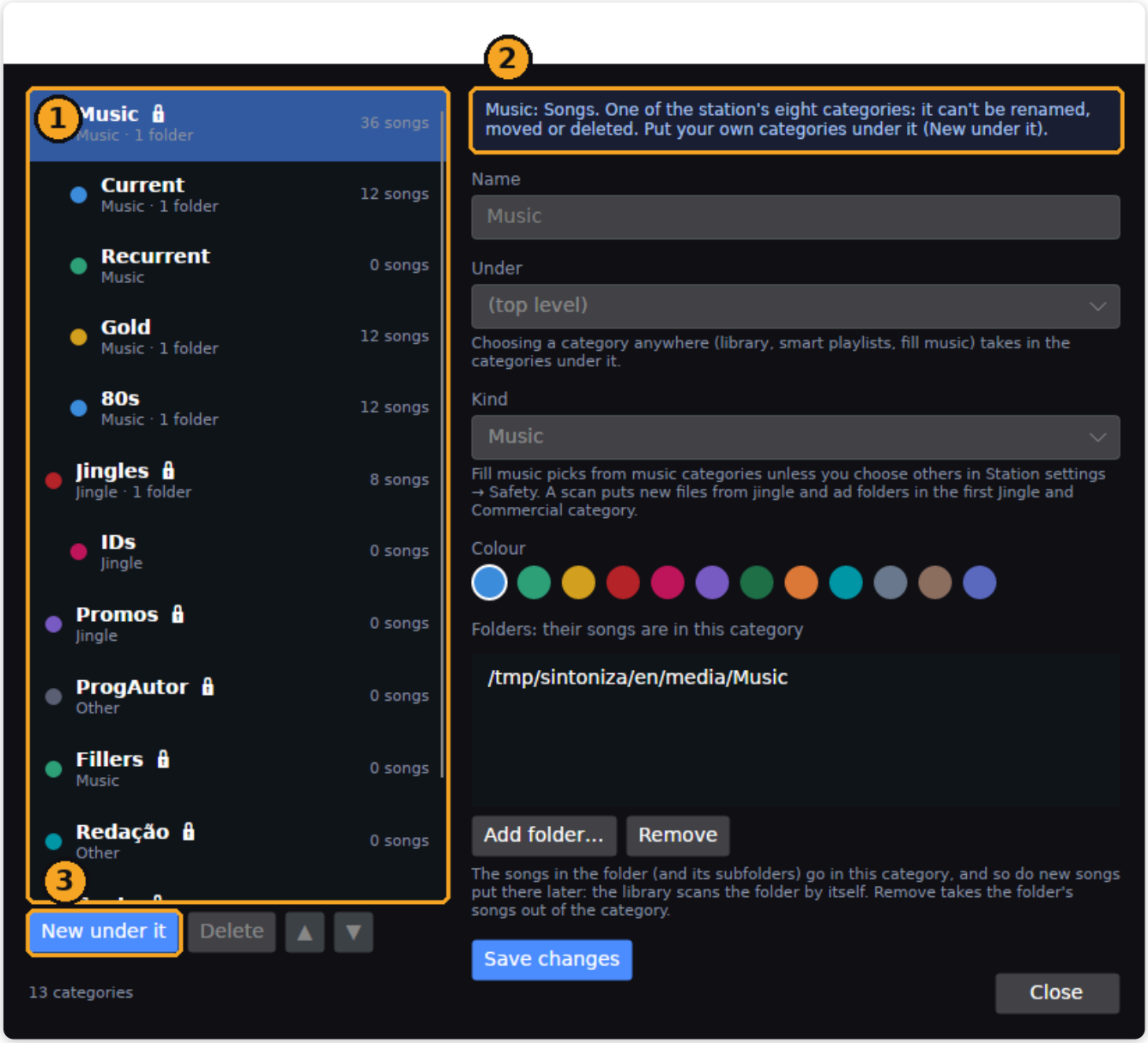
Sintoniza Pro: Home

1. The **menu**: File, Library, Programming, Traffic, Station, Help.
2. The **toolbar**: Save (Ctrl+S) and the tools used most.
3. The **tools**: Home; Library (songs, categories, import); Programming (hour templates, daily playlists, events); Traffic (ad breaks, clients and campaigns); Station (settings, cartwalls).
4. The clock and the date.
5. **Today**: the hours planned, ad breaks and adverts booked, events, voice tracks.
6. **Needs attention**: missing files, songs without a category, hours without a template, campaigns ending this week, unsaved changes.
7. **This computer**: system, processor, memory, free disk, how long it has been on. Next to it, the library and the database.
8. **Now playing** on air (from every studio's play history), and this hour's template.
9. The station, the database and who is working here.
10. The **status bar**: version, station, database, user, unsaved changes…
11. …and the date and time.

Categories

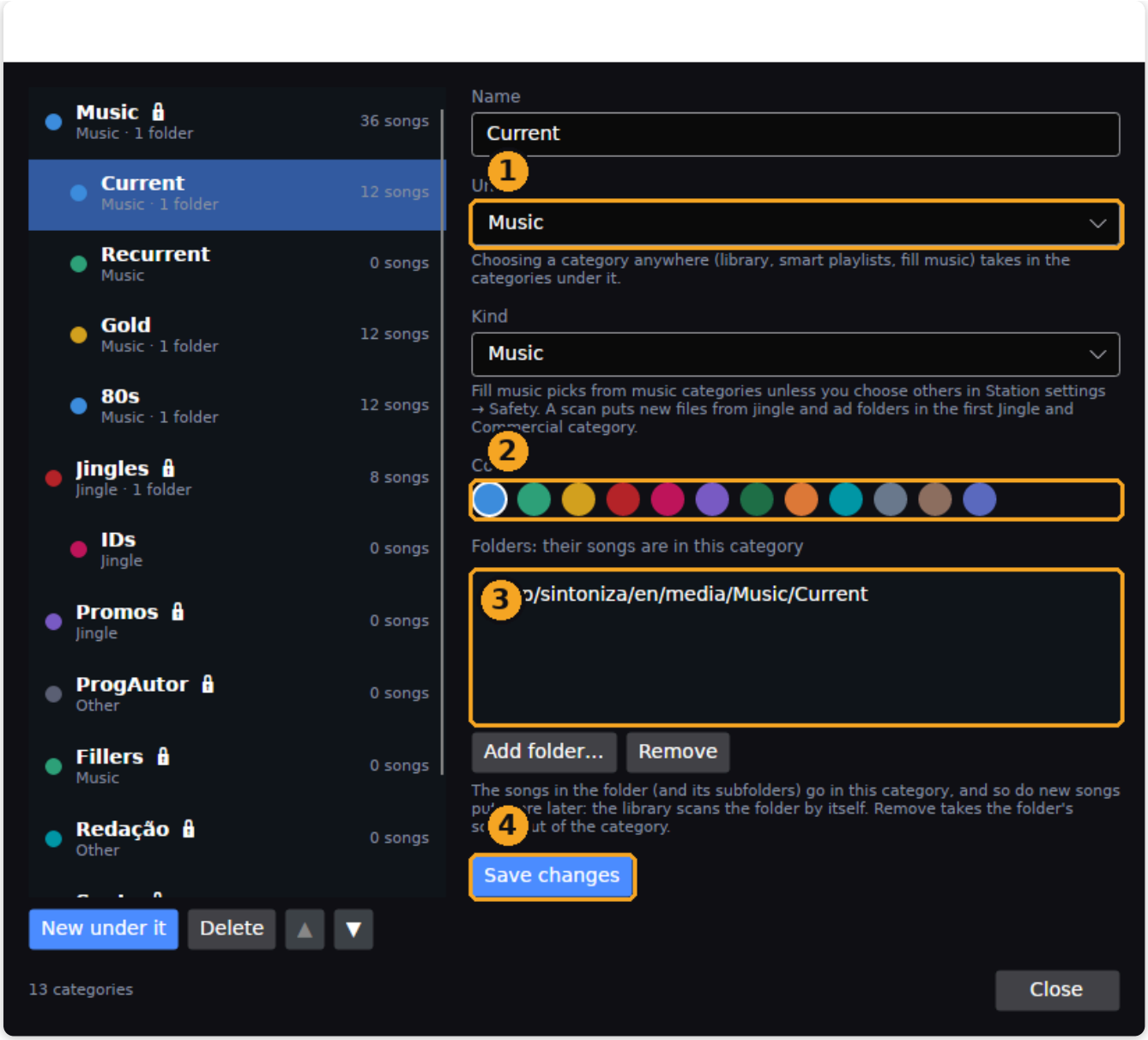
Every station has the same eight categories at the top, which can't be renamed, moved or deleted: **Music**, **Jingles**, **Promos**, **ProgAutor** (shows recorded outside), **Fillers**, **Redação**

(news), **Spots** (adverts) and **Takes** (voice inserts: the time, the temperature… as if a host were there). Your own go under them.



Categories: one of the station's eight

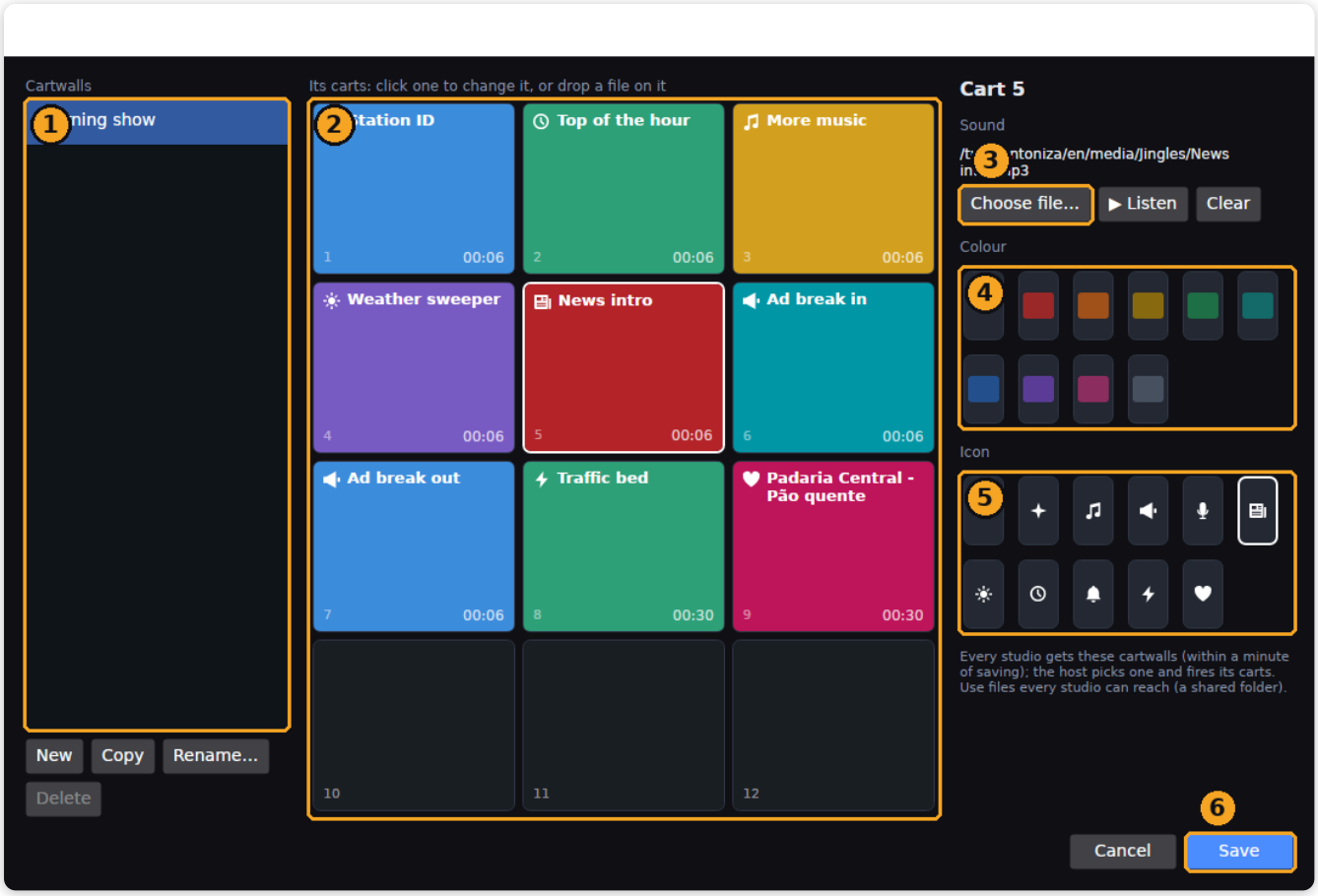
1. The categories, as a tree; 🔒 = one of the station's eight.
2. What it is for; only its colour and folders change.
3. **New under it**: a category of your own under the selected one (Music › 80s, Spots › Clients…).



Categories: a category of your own

1. **Under**: move it inside another category.
2. Its colour in the lists.
3. Its **folders**: their songs are in it, and new songs put there later too.
4. **Save changes**.

Cartwalls



Cartwalls

1. The cartwalls: **New**, **Copy**, **Rename…**, **Delete**.
2. Its 12 carts, as the studio sees them: click one to change it, or drop a file on it.
3. The cart's sound: **Choose file…**, **Listen** (Preview output), **Clear**.
4. Its colour…
5. …and icon.
6. **Save**: every studio gets them within a minute.

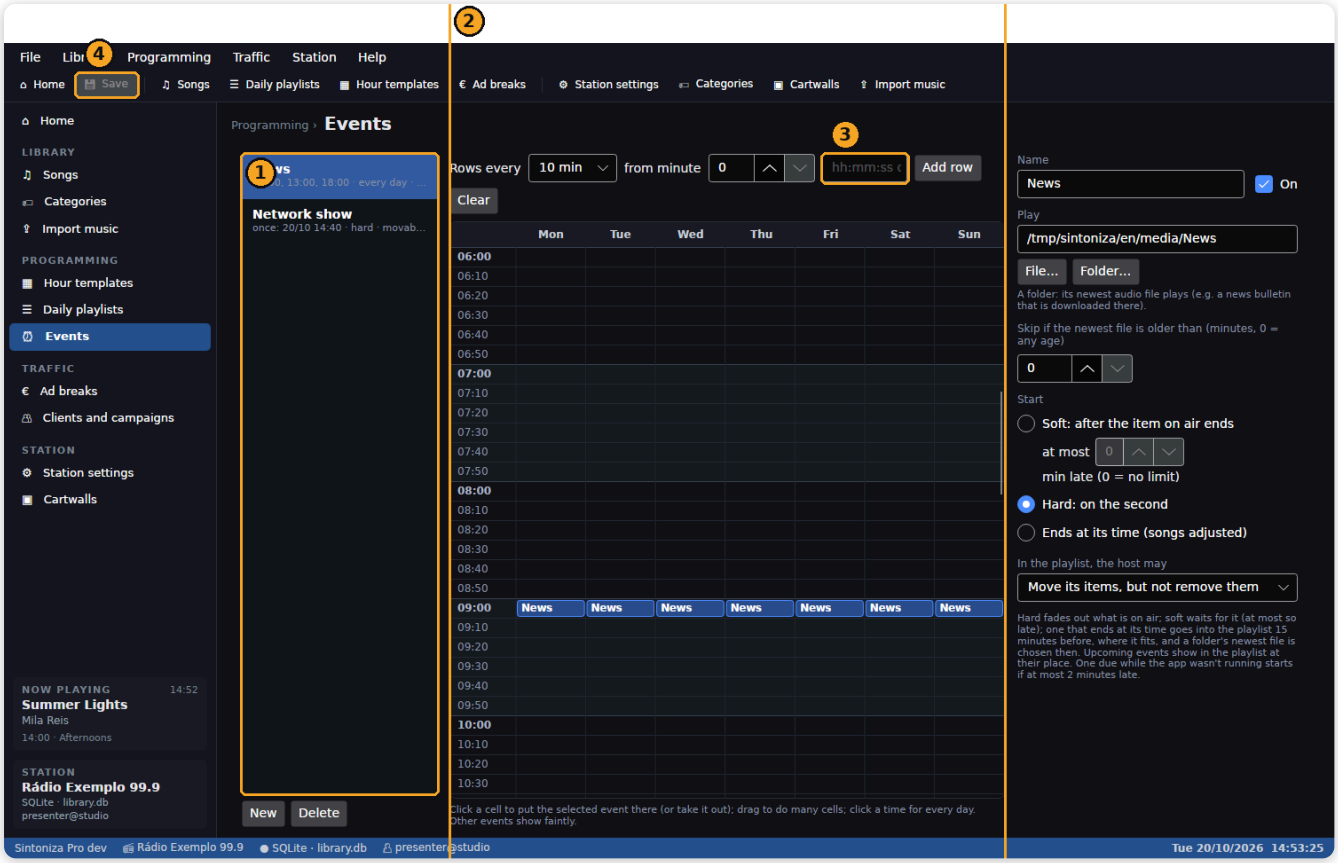
> **TIP**
> Use files every studio can reach (a shared folder), as for the music.

The schedule

Everything planned ahead: Events, Ad breaks, Clients and campaigns, Hour templates and Daily playlists, chosen on the left. **Save** (in the toolbar, lit when something changed, or Ctrl+S) writes it to the database and the on-air program takes it within a few seconds. If someone else saved meanwhile, you are asked before theirs is replaced.

> **TIP**
> **The week calendars** (Events, Ad breaks, Playlists): **click a cell** to place the selected item at that day and time (again to take it out), **drag** to do many cells, **click a time** for every day. **Add row** adds an exact time, e.g. `12:59:55`; `*:59:55` is that minute of every hour.

Events

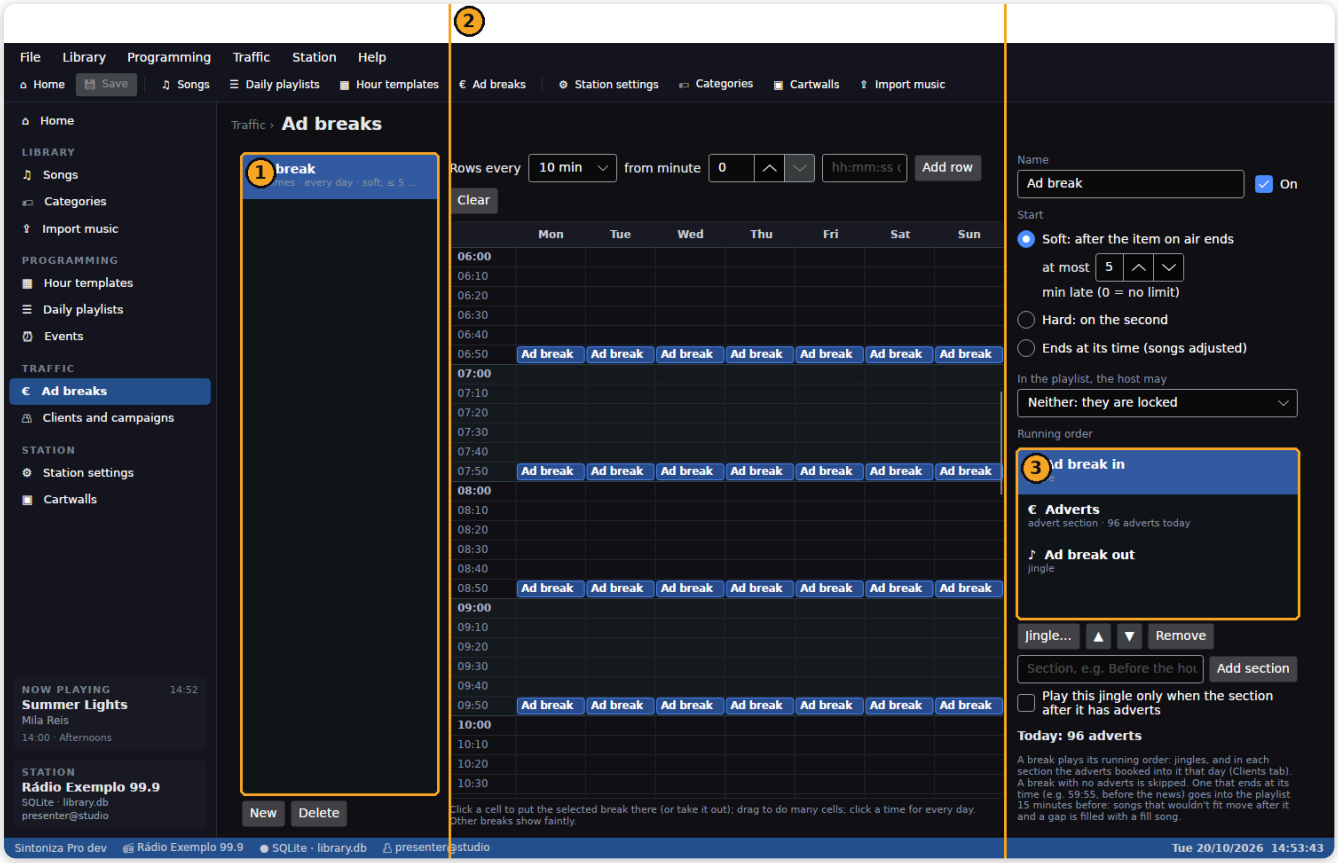


Events: things that play at set times

1. The events. **New** / **Delete** below.
2. The week: where the selected event plays.
3. An exact time to add as a row.
4. **Save** the whole schedule.

An event plays a **file**, the **newest file in a folder** (the news: skipped if older than N minutes) or an internet **stream** for N minutes. How it starts: **Soft** (after the item on air ends, at most N minutes late), **Hard** (on the second; what is on air fades out) or **Ends at its time** (the songs before it are adjusted). And what the host may do with it in the playlist: remove and move, only move, or nothing.

Ad breaks

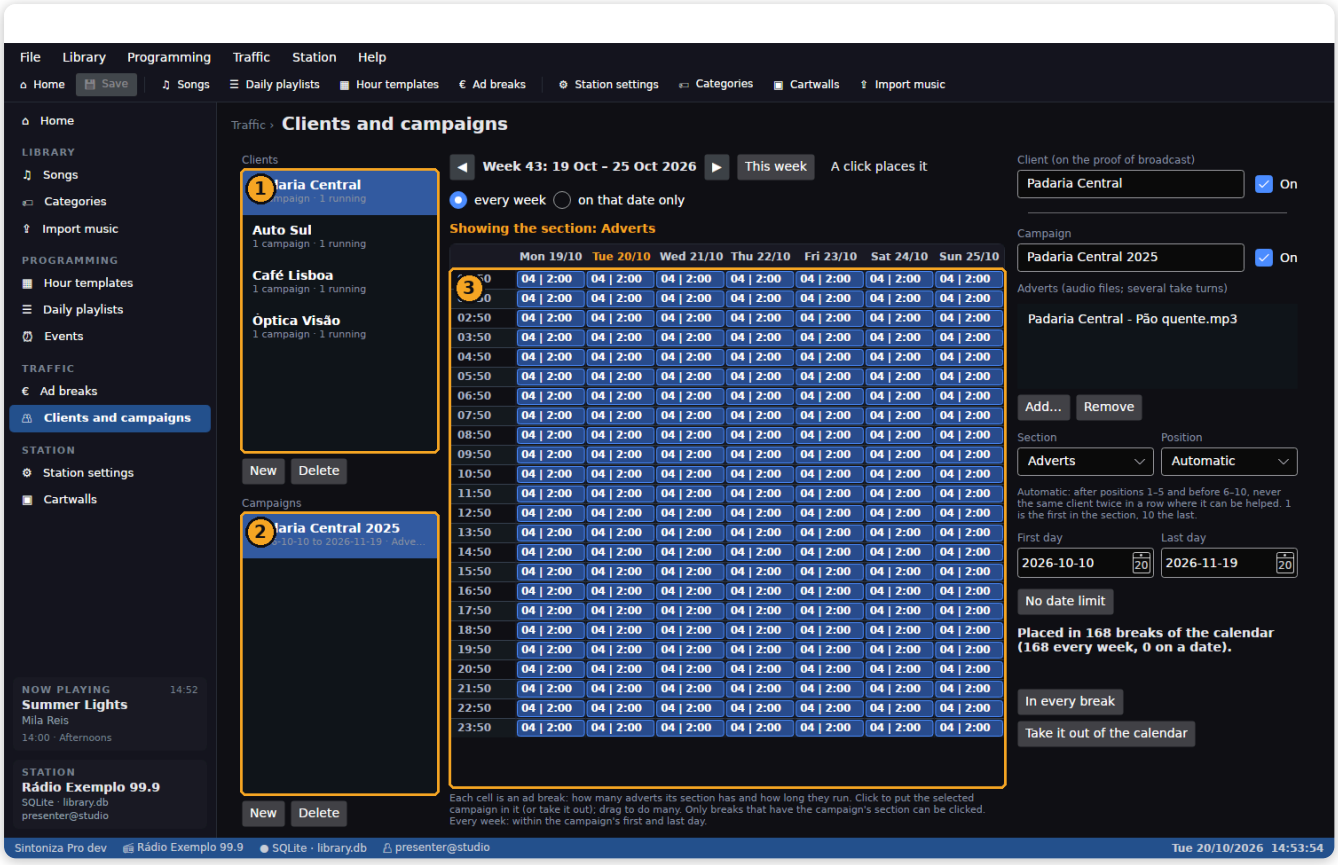


Ad breaks

1. The breaks.
2. Where they are in the week (here every hour at :50).
3. The **running order**: jingles and **sections**; the adverts booked in a section play there.

A jingle can be marked *play only when the section after it has adverts* (a section opener). A break with nothing booked is skipped. Breaks are locked in the playlist by default.

Clients and campaigns

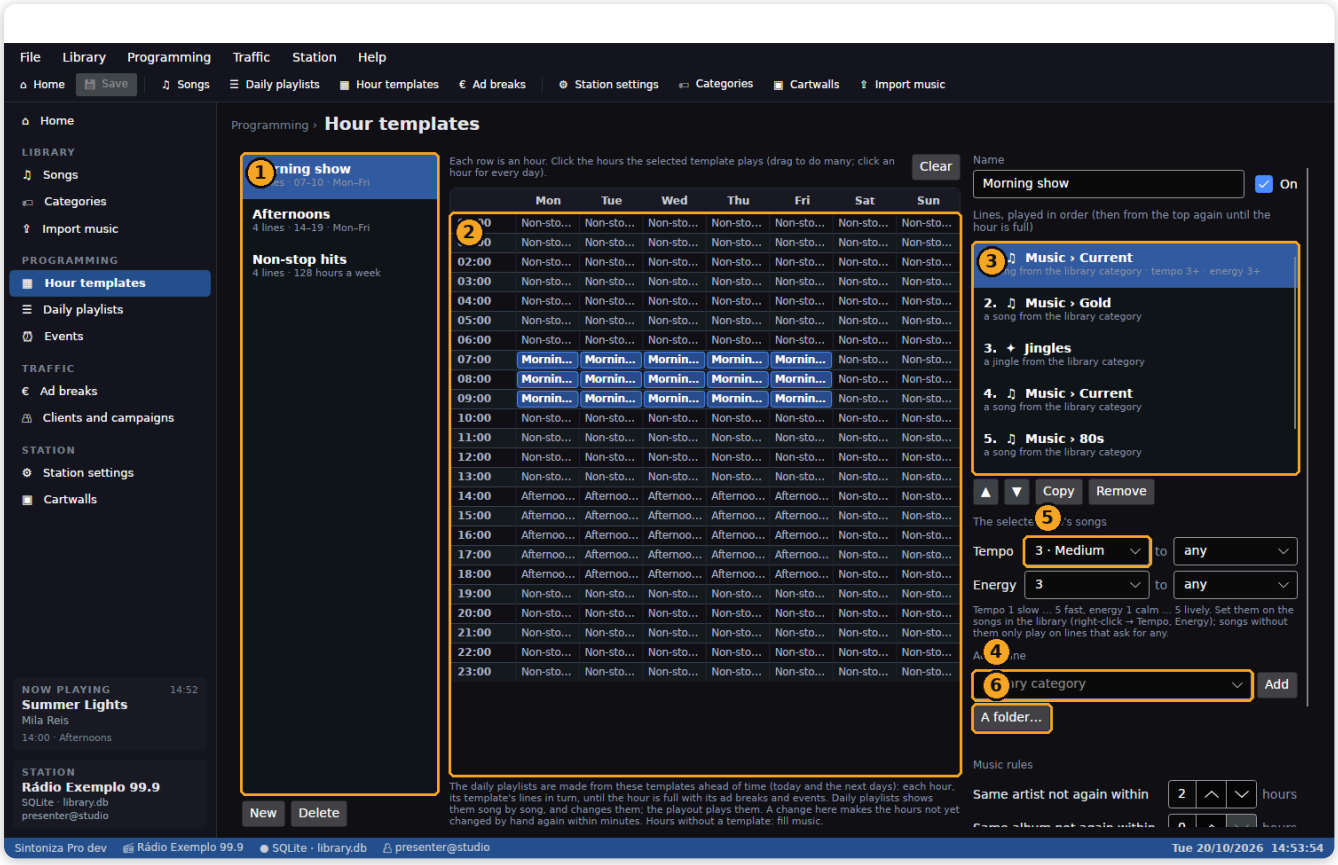


Clients

1. The **clients** (the name on the proof of broadcast).
2. Their **campaigns**: the adverts (several take turns), section, position, first and last day.
3. The week's ad breaks: how many adverts and how long each runs. **Click** a break to book the campaign there, every week or on that date only.

In every break books a campaign in every break that has its section. **Position**: automatic, 1–5 from the start of the section, or 6–10 at its end.

Playlists: hour templates for your shows

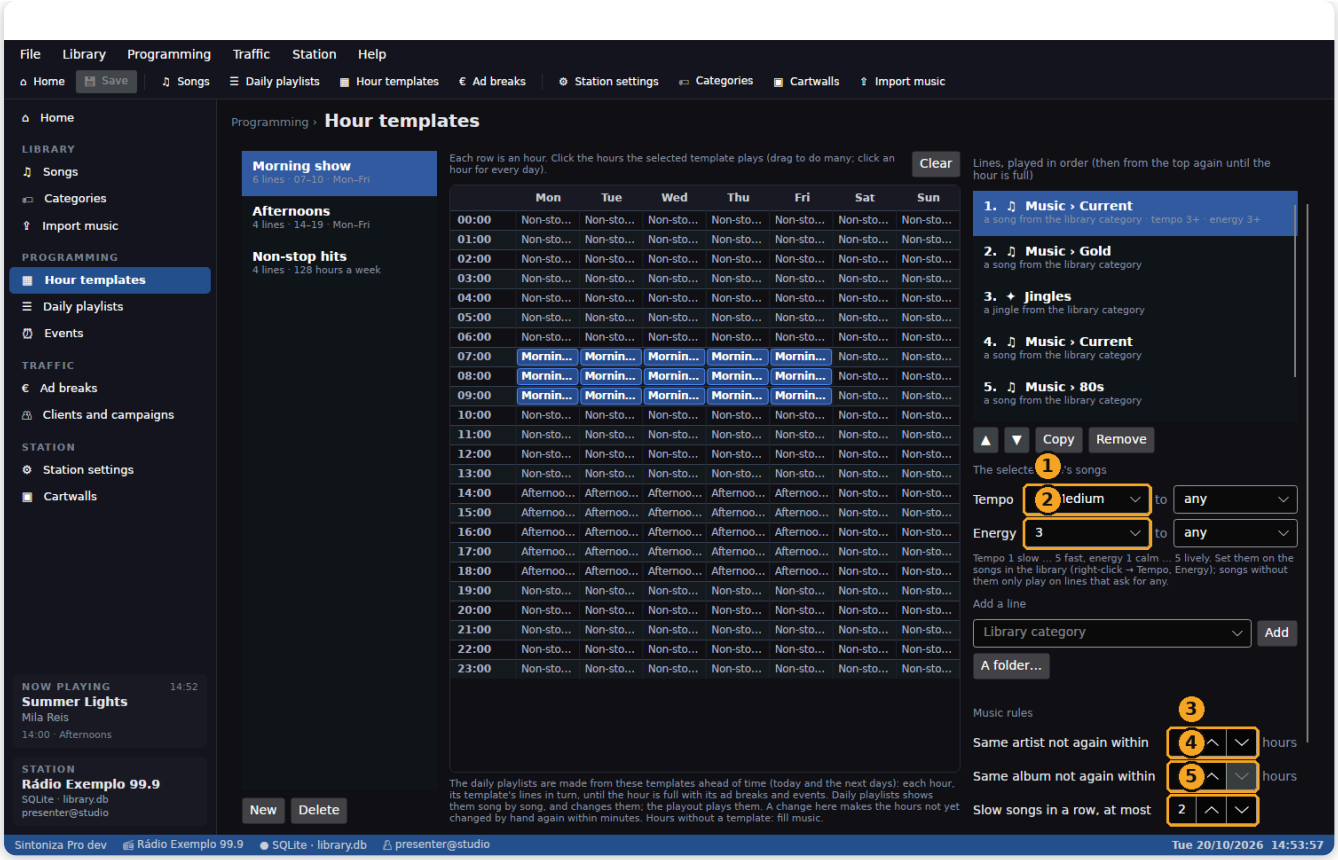


Hour templates

1. The templates: one per show or kind of hour.
2. The hours each plays: here the Morning show on weekdays 07–10.
3. Its **lines**, played in order and then from the top again until the hour is full.
4. A line is a song from a **library category** (Music › Current, Gold, Jingles…)…
5. The selected line's songs: a **tempo** and **energy** range (here the first song of the morning is lively).
6. …or a file from a **folder**.

Music rules

Give the songs a **tempo** (1 slow … 5 fast) and an **energy** (1 calm … 5 lively) in the library: select them, right-click → *Tempo* or *Energy*. Then a template's lines can ask for them, and the template keeps artists and albums apart.



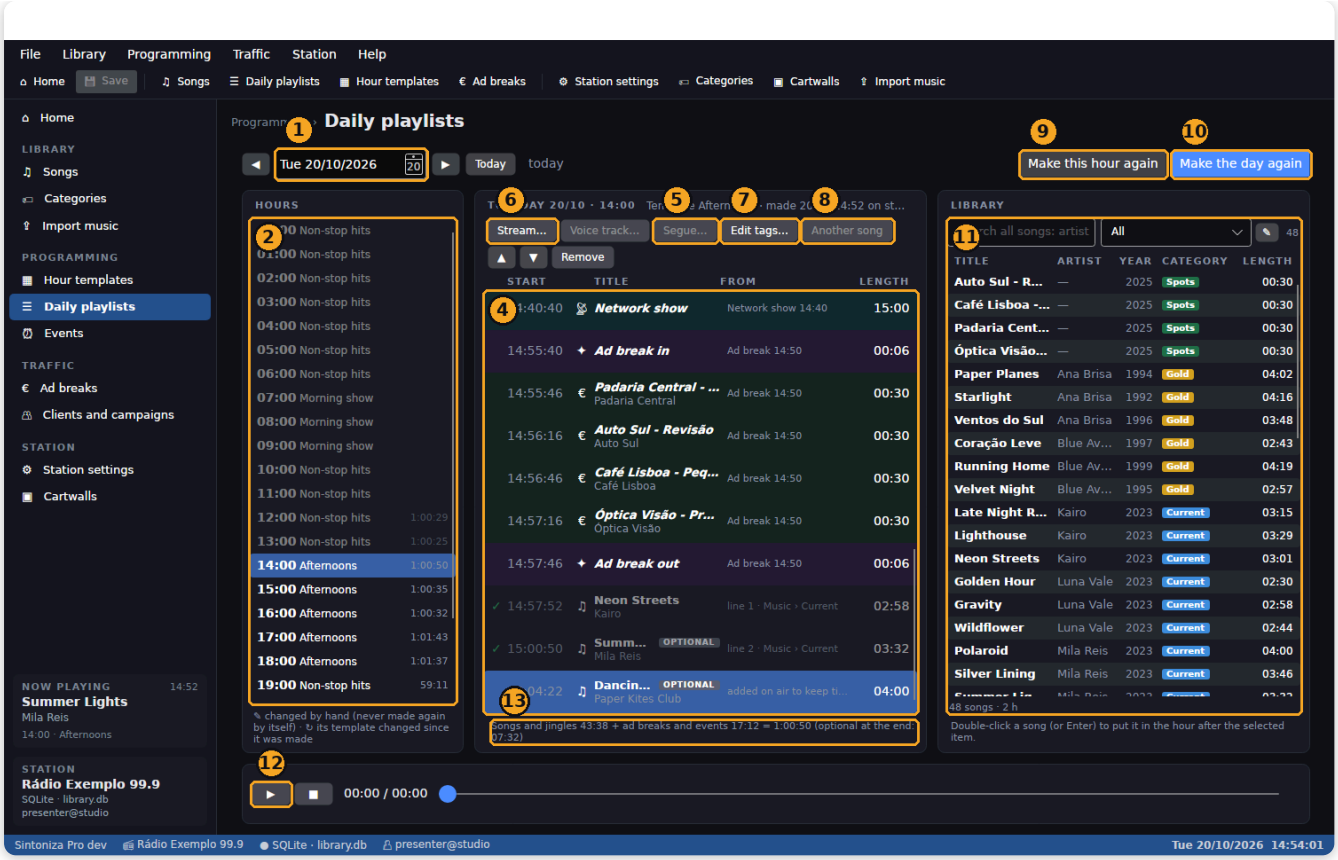
A template's music rules

1. **Tempo** from … to … for the selected line (*any* = no limit).
2. **Energy** from … to ….
3. The same **artist** not again within these hours (0 = the rest time in Settings → Safety).
4. The same **album** not again within these hours (0 = no album rule).
5. At most this many **slow songs** (tempo 1–2) in a row (0 = no limit).

> **TIP**
>
> A rule that would leave nothing to play is let go for that song (album first, then slow songs, then artist), so the hour is always full. A line whose tempo or energy no song has plays from its whole category, and Daily playlists says so.

Daily playlists

The hours as they will play, made from the templates: every song, jingle and advert, in order, with its start time.



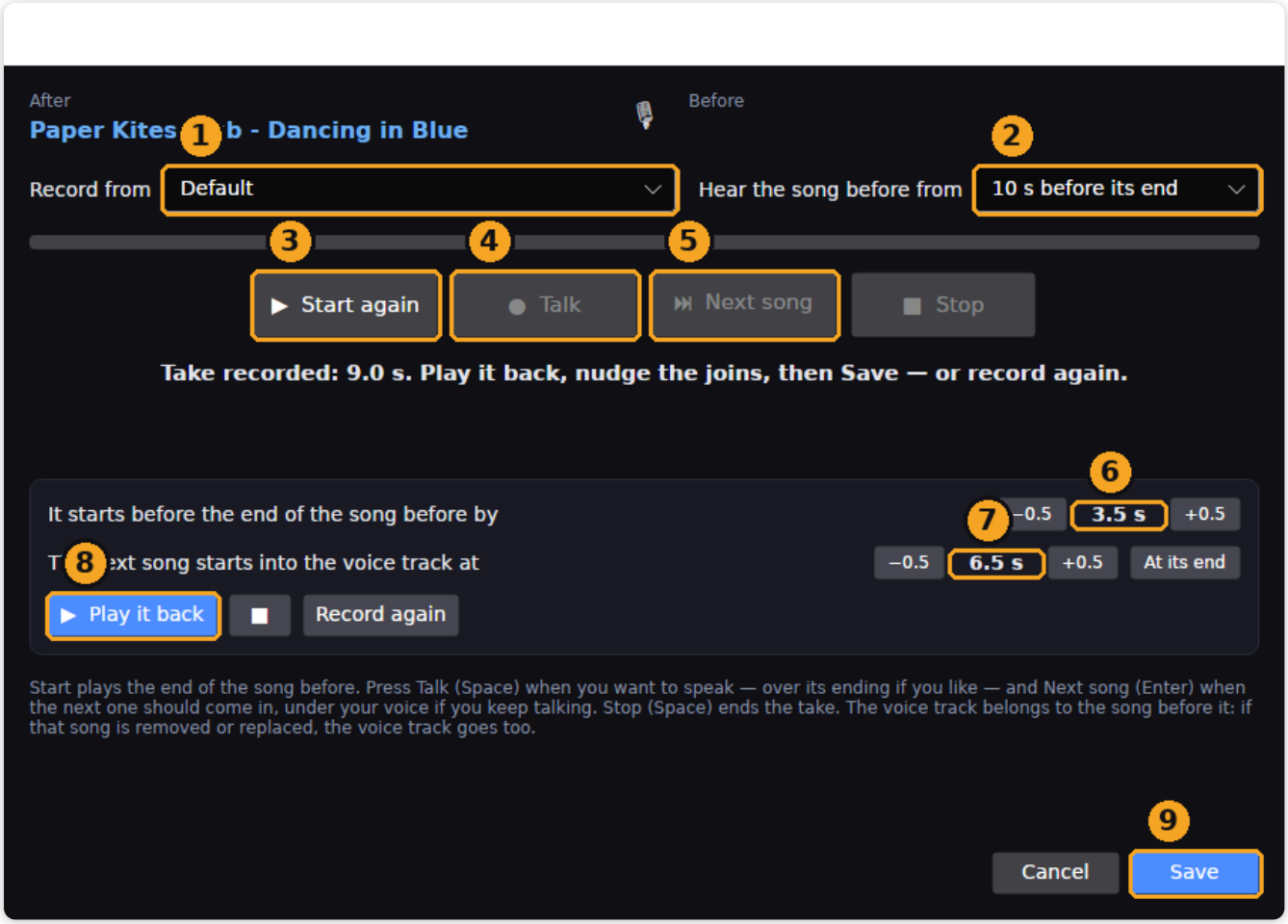
Daily playlists

1. The day: ◀ ▶ and **Today**.
2. Its 24 hours, with the template and length of each. ✎ = changed by hand; ↻ = its template changed since.
4. The hour: songs, jingles, the ad break's adverts (with the client) and events, each with its start time and where it came from. ✓ = played; ▶ = on air; OPTIONAL = only if there is time.
5. **Segue…**: how the selected song goes into the next (see Segues).
6. **Stream…**: another station's stream at a time in this hour, on the dot or after the item on air. Remove takes it out.
7. **Edit tags…** of the selected song.
8. **Another song** from the same template line; ▲ ▼ move, **Remove** takes it out.
9. **Make this hour again** from its template.
10. **Make the day again** (hours changed by hand are kept).
11. The library: **double-click** a song to put it after the selected item; right-click → Listen.
12. **Listen** to the selected item (or double-click it, or Space), on the Preview output only.
13. The hour's total: songs and jingles, ad breaks and events, and what is optional.

- **Build it song by song** with Another song, ▲ ▼, Remove and the library. Each change is saved at once and the hour is marked ✎; it is never made again by itself.
- **Or let the rules do it**: today and the next days are always made, and an hour whose template changes is made again within minutes (unless it was changed by hand or is about to play).
- The rules: the songs rested longest first; a song not again within N hours, an artist not within N minutes (Station settings → Safety); never the same artist twice in a row while another can play.
- Songs the on-air program adds to keep time appear as *added on air to keep time*.

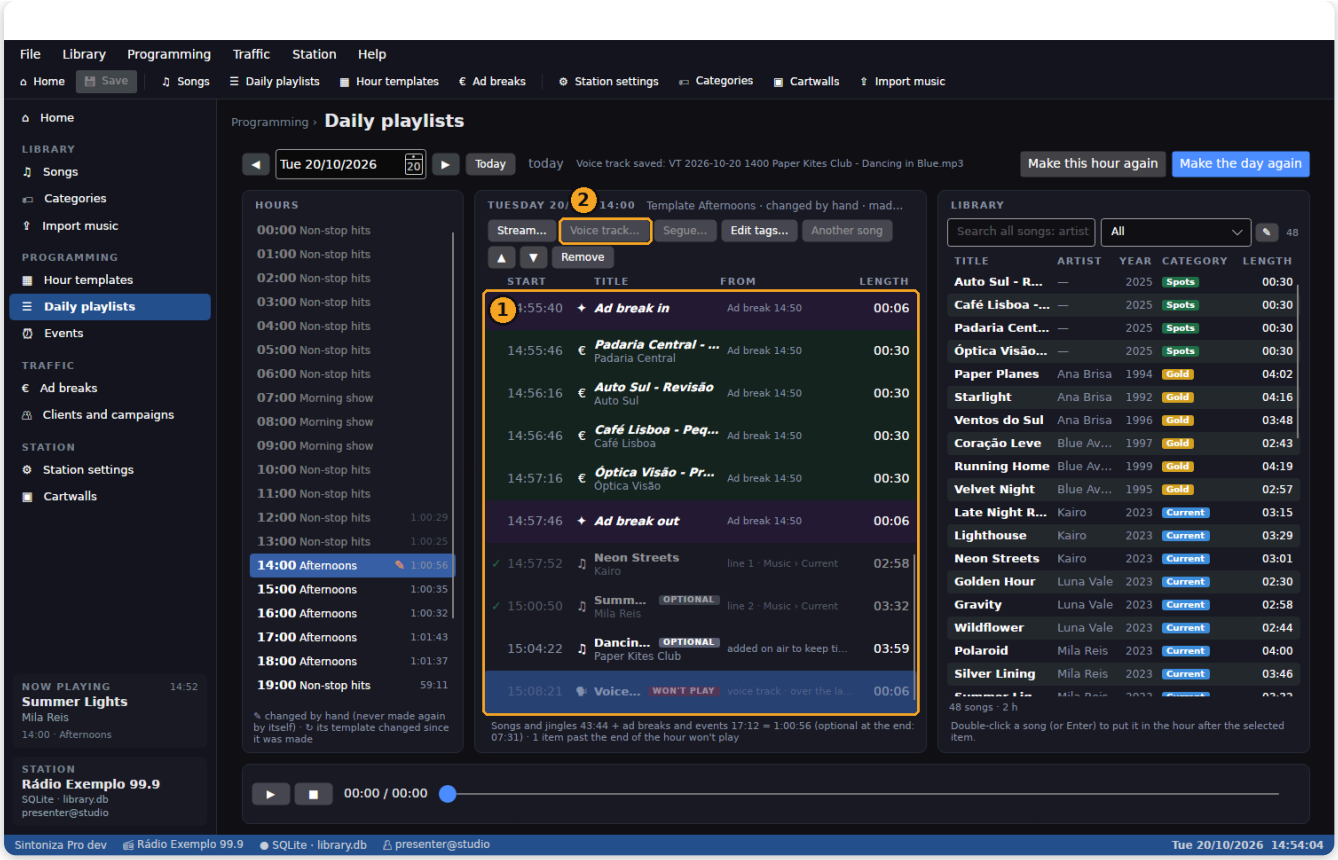
Voice tracks

Record your talk between two songs ahead of time, as if you were live: select a song in an hour and click **Voice track…**.



Voice track…

1. The **input** to record from (the studio mic, or any input of this computer).
2. How much of the song's end you hear first.
3. **Start** plays the end of the song before, on the Preview output.
4. **Talk** (Space) starts recording — over the song's ending if you like; again to stop.
5. **Next song** (Enter) brings in the following item while you talk, under your voice.
6. After the take: how far it starts over the song's end…
7. …and where the next song comes in. ±0.5 s to nudge them.
8. **Play it back** as on air; **Record again** if you like.
9. **Save**: kept as an MP3 in the station's voice tracks folder and in the library under **Takes**.



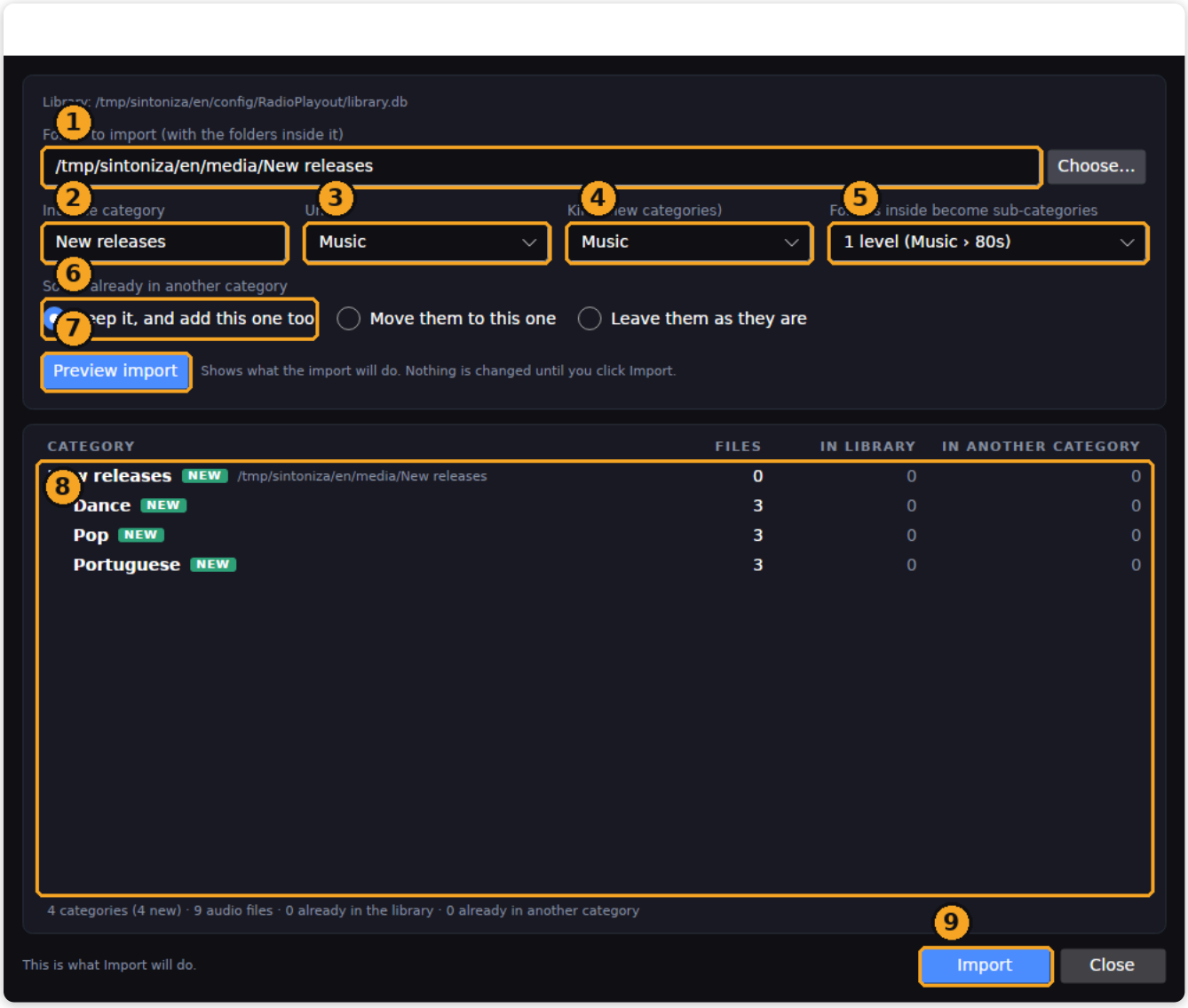
The voice track in the hour

1. It goes right after its song (🗣), starting over the song's last seconds.
2. **Voice track...** again on the song, or on the voice track, to record it again.

> **TIP**
>
> A voice track belongs to the song before it: removing or replacing that song takes the voice track too, moving the song moves it, and on air it is skipped if the song can't play.

5 Sintoniza Import

Brings a folder of music into the library, with the folders inside it as categories.



Sintoniza Import

1. The **folder** to import (**Choose…** next to it).
2. The **category** it goes into (by default the folder's name)…
3. …**under** one of the station's eight (Music by default) or a category under them.
4. The **kind** of new categories: it follows the one they go under.
5. **Folders inside become sub-categories**: none, or 1–3 levels (Music/80s → Music › 80s).
6. Songs already in another category: keep it and add this one too, move them, or leave them.
7. **Preview import**: what it will do. Nothing is changed yet.
8. The categories it will make (NEW or ALREADY THERE), the files in each, how many are in the library already and how many are in another category.
9. **Import**: makes the categories, reads the songs and puts each in its category.

Each folder is kept for its category: songs put there later get it too (the on-air program scans those folders by itself). **Stop** halts an import; importing again finishes it.

radio-import (command line)

The same scan without a window, e.g. every night from cron or Task Scheduler:

```
radio-import                          scan the library's folders
radio-import /music /jingles          scan these folders
radio-import --db /path/library.db    an SQLite library file
radio-import --mysql user:password@server[:port]/database
radio-import --stats                  what is in the library
```

Only new and changed files are read. Deleted files, and the songs of a deleted folder, are taken out; a drive or share that isn't connected is left alone.

6 How do I…

…set up a weekday morning show?

1. Pro → **Playlists** → **New**; name it.
2. Add its lines: e.g. Music › Current, Gold, Jingles, Current, 80s.
3. Drag over the hours 07–10, Monday to Friday. **Save**.
4. Check it on **Daily playlists**: the hours are made within minutes.

…put the adverts last in the hour?

1. Pro → **Ad breaks** → **New**; give it a running order with a section.
2. **Add row** `*:59:00` and click that time for every day.
3. Choose **Ends at its time**, so it ends on the hour.
4. Book the campaigns on the **Clients** tab. **Save**.

…change a song in tomorrow's 09:00?

1. Pro → **Daily playlists** → ▶ to tomorrow → **09:00**.
2. Select the song → **Another song**, or **Remove** it and double-click one in the library.

…put a folder of songs in Gold?

Sintoniza Import with that folder, *into the category* Gold, no sub-categories. Or in the on-air program: Folders tab → right-click the folder → Category → Gold.

…get the music report for the collecting society?

On-air program → **Reports** → *Music played (royalties)* → the month → **Save as CSV…**.

…send an advertiser the proof of broadcast?

Reports → *Adverts (proof of broadcast)* → the advertiser and the dates → **Save as a page to print…** → print to PDF.

7 When something isn't right

What you see	What to do
The on-air playlist doesn't match Daily playlists	Press **Clear** in the on-air program: it refills from the plan, from the item planned for that moment. If it runs ahead or behind, the red line on Daily playlists says by how much; it catches up by the next hour.
Only jingles and adverts play	The songs' files moved or were deleted: fix the categories' folders (Pro → Categories) and click **Scan** in the Library tab.
A song is skipped with “FileOpen”	The file isn't where the library says: **Scan** again.
Listening is heard on air	Settings → Audio → **Preview**: choose an output that isn't on air (headphones, PFL).
Something went wrong on air	Settings → Safety → **Open log folder**: one file a day. Settings → Alerts can warn you by Telegram or e-mail.